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SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING

PPG INDUSTRY PROJECT

TITLE : XXX

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In modern manufacturing, the accumulation of Recoverable Assets poses a silent but significant threat to financial stability, specifically raw materials exceeding months of consumption. When such materials are unmanaged well, they often transition into dead stock or expire, meeting the Magna Carta criteria for inventory obsolescence [1]. Without an automated system to track these items, organizations face “blind financial risks” as they cannot accurately quantify the exact amounts required for finance. This lack of transparency leads to inflated assets valuations on balance sheets and inefficient use of warehouse capacity [2].

A critical disconnect exists between inventory levels and customer fulfillment. Organizations may experience excess inventory for certain materials while facing outstock risks for others due to demand variability and lead time fluctuations[3]. Such inefficiencies are frequently driven by fragmented data systems and the absence of integrated data pipelines, which hinder real-time decision making and increase operational risk [2][3].

The assets and risks management are often undermined by data invisibility, where a lack of integrated pipelines leads to financial consequences. A well documented example is Cisco System inventory crisis, where a failure to distinguish between usable assets and obsolete stock during a market shift resulted in a massive \$2.25 billion write-down, proving that without a real-time “single source of truth”, excess inventory quickly becomes a permanent financial liability (*Source: Case Study: Cisco’s approach to Disaster Recovery and Risk Management*) [4]. Industry benchmarks indicate that inventory carrying costs typically range between 20% and 30% of total inventory value, while poor inventory visibility contributes to excess and obsolete stock accumulation, reducing operational efficiency [5].

1.2 Problem Background

The increasing accumulations of excess raw materials, often categorized as recoverable assets, presents a critical challenge in modern manufacturing systems. Materials that exceed projected consumption levels are highly susceptible to becoming obsolete or transitioning into dead stock when not properly managed. This issue is intensified by the lack of inventory

visibility, where organizations do not possess real-time insights into stock conditions, usage patterns, and lifecycle status. As a result, companies struggle to accurately identify surplus inventory or detect materials at risk of obsolescence, leading to inefficient utilization of resources and increased operational costs [2][3].

Moreover, the absence of integrated data systems creates a fragmented information environment, preventing organizations from achieving a unified, end-to-end view of inventory across the supply chain. Without a centralized and automated data pipeline, inventory data remains siloed, limiting the ability to align procurement, production, and demand forecasting processes. This disconnection significantly elevates financial risk, as excess and obsolete inventory may go undetected until it necessitates costly write-downs [5].

1.3 Project Aim

The aim of this project is to design and develop an end-to-end data engineering pipeline for inventory risk management, focusing on the identification and monitoring of recoverable assets and obsolete inventory. The system aims to automatically detect raw materials exceeding defined consumption thresholds, classify items at risk of becoming dead stock, and quantify potential financial exposure associated with excess and obsolete inventory.

Additionally, this project seeks to enhance inventory visibility by integrating fragmented data sources into a unified, analytics-ready data model. Through the implementation of real-time dashboards and key performance indicators (KPIs), the proposed solution aims to support data-driven decision-making, reduce inventory-related risks, and improve the alignment between supply, demand, and financial reporting within manufacturing operations.

1.4 Project Objectives

1. To provide a layered architecture (Bronze, Silver, and Gold Layers) for end-to-end data engineering pipelines for inventory risk management.
2. To automatically identify and categorize inventory risk, including Recoverable Assets (RA), dead stock, expired supplies, and stockout hazards.
3. To increase data visibility and consistency by combining several dispersed data sources into a single data platform.

4. To put ETL/ELT procedures into place to guarantee that data is appropriately transformed, cleansed, and stored for analysis.
5. To create dynamic dashboards that facilitate real-time monitoring and decision-making by utilizing business information tools.
6. To calculate the possible financial risk connected to excess and outdated inventories.

1.5 Project Scope

The development of a data-driven inventory risk management system in a manufacturing setting is the main goal of this project. The project's scope comprises:

1. Data Integration
 - Gathering and combining inventory information from several sources into a system for central storage.
2. Data Engineering Pipeline
 - Managing raw, processed, and analytic-ready data by creating a three-layer architecture (Bronze, Silver, and Gold layers)[6]
3. Data Processing (ETL/ELT)
 - Performing data extraction, transformation, and loading process to ensure data quality and usability for analysis.
4. Inventory Risk Identification
 - Utilizing business principles to identify stockout concerns, dead stock, expired commodities, and recoverable assets based on inventory conditions.
5. Cloud-based Implementation
 - Storing, processing, and managing data with Microsoft Azure services.
6. Data Visualization
 - Creating Power BI dashboards to support decision-making and provide important information[6].
7. limitations

- This research primarily concentrates on rule-based risk identification and excludes machine learning models and sophisticated predictive analytics.

1.6 Project Importance

This project is significant because it tackles major issues with inventory control and financial risk in the contemporary industrial system. First, by combining disparate data into a consolidated system, the initiative enhances inventory visibility, enabling businesses to more effectively manage stock levels and conditions[7][8]. Second, it makes proactive risk management possible by spotting dead stock, excess inventory, and expired goods before they cause large financial losses[7]. Furthermore, compared to manual inventory tracking techniques, the use of an automated data engineering pipeline lowers human error and increases data accuracy [8]. Additionally, by displaying insights through interactive dashboards and key performance indicators (KPIs), data visualization technologies improve data-driven decision-making [6]. Lastly, by optimizing stock management and matching supply and demand, this project helps lower inventory-related expenses and increase operational efficiency, ultimately improving overall business performance [7][8].

1.7 Report Organization

This report is structured into 5 chapters to provide an overview of this project

Chapter 1 : Introduction

This chapter provides the background of the project, along with the project's overall aim, specific objective, scope, and importance to modern manufacturing

Chapter 2 : Literature Review

This chapter discuss the fundamental theories of data engineering pipelines and inventory management concepts, reviews related previous research, and the details of the core technologies that will be utilized, such as Microsoft Azure and Power BI

Chapter 3 : Methodology

This Chapter Discusses the chosen project methodology, outline the specific phases of development, and justifies the technology stack used to build the solution

Chapter 4 : Result and discussion

This chapter present the details the system architecture, requirement analysis, database design, and interface design required to structure the three-layer data pipeline and visualization dashboard

Chapter 5 : Conclusion and Recommendation

This chapter summarizes the overall project outcomes, evaluates the successful achievement of the project objectives, and provides recommendations for future improvements.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter is about the key concept ,systems,and technologies related to the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system.This chapter provides a theoretical theory for the development of the proposed system, including supply chain management, Enterprise Resource Planning (ERP) systems, data engineering pipelines based on the Medallion architecture, ETL/ELT processes, inventory management principles, and data visualization techniques.

In addition,this chapter also reviews prior studies and existing systems related to inventory analytics and data-driven solutions.A comparison of the system is conducted to identify respective strengths and limitations.This evaluation highlight current gaps and support the rationale for the proposed system.Futhermore,the technologies used in this project such as Microsoft Azure and Power BI are also discussed in this chapter to demonstrate their role in enabling efficient data processing,storage,and visualization.

Overall,this literature review forms the theory and technological foundation needed to design and develop the proposed inventory risk management system.

2.2 Fundamental Theory & Concept

This section explains the fundamental theories and concepts related to data engineering, inventory management and data visualization of the proposed system.

2.2.1 Data Engineering Pipeline

A data engineering pipeline refers to a series of processes used to collect, transform and store data for analysis. It ensures that raw data is converted into meaningful and usable information for decision-making. In this project, a three-layer data pipeline is used:

- Bronze Layer (Raw Data)

This layer stores raw data from multiple sources into a data lake. Basic data quality checks such as null detection, duplicate data removal and data type validation are performed.

- Silver Layer (Processed Data)

Data is cleaned and transformed in this layer. Business logic starts to be applied to identify inventory conditions such as excess stock, dead stock and expired materials.

- Gold Layer (Analytics- Ready Data)

This layer contains structured and optimized data, typically organized into data warehouse schema. It is used for reporting and visualization through tools like Power BI.

The layered data pipeline improves data quality, scalability and maintainability of the system.

2.2.2 ETL/ELT Process

ETL means Extract, Transform, Load, which is a process used in data integration. Extract is the process of collecting data from various sources such as CSV files. Transform is a process of cleaning, filtering and process according to business rules. Load means transformed data is stored in a data warehouse for analysis.

In modern cloud systems like Microsoft Azure, the ELT approach is often used, where data is first loaded into storage and then transformed. This approach improves performance and scalability.

2.2.3 Inventory Management Concepts

Inventory management is a process of monitoring and controlling stock levels to ensure efficient operations and minimize losses.

Key concepts used in this project include:

- Recoverable Assets (RA)

These are materials where stock levels exceed 12 months of consumption. This indicates overstocking and potential financial inefficiency.

- Magna Carta Dead Stock

Materials that have not been consumed for more than 9 months and have no expiry date. These items are unlikely to be used and may result in financial loss.

- Expired Materials

Materials that have passed their expiry date and can no longer be used. These require full financial write-down.

- Stockout Risk

This occurs when inventory levels are too low to meet demand, especially considering supplier lead time. It may result in delayed production or unfulfilled customer orders.

These concepts are essential for identifying risks and making better inventory decisions.

2.2.4 Data Visualization and Business Intelligence

Data visualization is the process of presenting data in graphical formats such as charts and dashboards. It helps users understand patterns, trends and insights more effectively.

Business Intelligence (BI) tools such as Microsoft Power BI are used to monitor inventory status, identify risks such as overstock and shortages, and support decision-making through interactive dashboards.

Visualization plays a critical role in transforming processed data into actionable insights.

2.3 Related Previous Researches or Systems / Compare Between Existing Systems / Ensure Strength and Weakness Between Systems.

In the modern industrial landscape, inventory management has transitioned from a simple logistical task to a strategic pillar of supply chain resilience. Historically, companies centered their management on balancing replenishment lead times (RLT) and carrying costs using static mathematical models. However, the rise of global market disruptions has underscored a need to move away from these traditional methods towards “intelligent” data-driven approaches. This transformation is particularly critical for large-scale operations, such as those at PPG, where the failure to accurately detect material obsolescence can lead to significant financial write-downs and operational inefficiencies.

Research by Liu et al. (2025) highlights that while past sales data is a “critical element” in predicting demand and preventing stockouts, relying on this data in isolation often results in forecasting inaccuracies. Their study emphasizes that modern systems must integrate multiple

features including current inventory levels and short-term demand forecasts to handle the complexities of large-scale, “imbalanced” datasets where critical risk events are rare but impactful. This finding supports the necessity of the “Silver Layer” in the proposed PPG pipeline, which goes beyond simple sales tracking to apply complex business logic for flagging “Recoverable Assets”.

Furthermore, Mulaka et al. (2025) demonstrate that traditional inventory practices, which often rely on manual sampling-based counting, are; “labor-intensive, time-consuming, and susceptible to human error”. Their work on automated forest nursery systems proves that moving toward real-time, automated inventory systems provides more “reliable data, directly improving decision-making” and lowering operational costs. For a chemical industry leader like PPG, this automation is essential to eliminate the human error associated with manually tracking thousands of “lot expiry dates” across millions of data rows.

Finally, the financial implications of inventory failure are detailed by Gravier and Swartz (2009). Their research identifies obsolescence as a major “life cycle cost driver” that can increase total organizational costs by 5% to 40% more than the optimal amount. They strongly advocate for "proactive obsolescence management strategies” rather than reactive ones to minimize the “large expenses” required to maintain outdated stock. This perspective provides a direct academic foundation for the PPG project’s goal of quantifying “Magna Carta provisions” –ensuring that financial teams can proactively account for write-downs before they escalate into significant losses.

To ensure the strength of the proposed Azure pipeline, it is necessary to compare it against the systems traditionally used in the industry. The following table contrasts manual processes and legacy Enterprise Resource Planning (ERP) reporting with modern data engineering.

Table 2.3.1 Comparison of Existing Systems

Feature	Traditional Manual (Excel)	Legacy ERP Reporting (SAP/Oracle)	Proposed Azure Pipeline
Data Integrity	High risk of human	Moderate; often siloed	High; automated

	error.	data.	ingestion and cleaning.
Logic Complexity	Limited to basic formulas.	Rigid; hard to customize for specific rules.	Flexible; applies complex “Magna Carta” logic.
Operational Speed	Reactive; slow updates.	Periodic batch updates.	Proactive; real-time analytics ready.
Financial Risk	High; 40% cost increases risk.	Low visibility into “dead stock.”	Low; automates “provision” calculations.

In summary, existing literature confirms that traditional inventory systems are often “reactive”, identifying obsolescence and stockouts only after they have impacted the bottom line. Previous research consistently points to manual processes as being prone to error and simple sales-based forecasting as being insufficient for modern risk management. By contrast, the proposed three-layer Azure pipeline represents the “proactive” management strategy recommended by industry experts. It integrates automation to reduce human error with the data intelligence required to manage the high-stakes financial risks associated with material expiry and the “Magna Carta” dead stock policy.

2.4 Technology Used

2.4.1 Microsoft Azure

Azure Data Factory

Azure Data Factory (ADF) acts as an automated delivery system and traffic controller for your entire project. ADF used automated workflows called “pipelines” to extract the data from the

resources and load it into the storage system [1]. ADF does not alter, clean or analyse the data itself. Its responsibility is to make sure the raw data is transported on schedule into bronze layer so the rest of the data pipeline can begin its work [2]

Azure Data Lake Storage

Azure Data Lake Storage is a massive and secure place that you can load your files. It's like a highly scalable storage that is designed to hold a massive amount of big data in its original format [3] . In this project, this will act as a Bronze Layer because it allows storing all of the raw CSV files securely. It acts as the foundation storage layer before any complex cleaning or processing start [4]

Azure Databricks

Azure Databricks is the heavy duty computing engine of the pipeline, serving as the place where all the actual data cleaning and complex calculations have been executed [5]. After pulling the raw data from the data lake, Databrick uses distributed cloud computing to transform it into a silver layer. In conclusion, Databrick turns the messy raw data into a highly valuable dataset by filtered, cleaned and refined [6].

Azure SQL Database

Azure SQL Database is needed to create perfectly structured data so that business reports can read it easily and quickly. This fully managed relational database acts as Gold layer [7]. These tools will organize the refined data into formatted “Star Schema,” which separates the data into fact tables and dimension tables. Instead of a vast lake of loose files, the SQL Database provides a highly organized, indexed, and instantly searchable environment that is efficient for analytics [8]

2.4.2 Microsoft Power BI

Microsoft Power BI serves as the visual representation layer that brings together all of your data work to life for business stakeholders [9]. Business managers do not want to look at the multiple rows of numbers in a database, instead they want to visualise the data where Power BI comes handy because it can translate those numbers into an interactive dashboard [10]. It allows us to build the specific graft, chart, and KPIs that are required to achieve the project goals [11].

2.5 Summary

This chapter presents a comprehensive overview of key concepts, theories and technologies essential for the development of the Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The discussion reveals highlights the importance of intergrating supply chain management, ERP systems, and modern data engineering approaches such as the Medallion architecture to improve inventory management and decision-making.

The discussion reveals how data engineering pipelines and ETL/ELT processes can be used to transform raw data into dependable and valuable information. Moreover, the concepts of inventory management mentioned, including Recoverable Assets, Magna Carta Dead Stock, expired materials, and stockout risk, show how data-driven solutions could be used to eliminate inefficiencies and financial losses.

In addition, the review of existing research and systems further reveals limitations in current inventory management approaches. For instance, reliance on historical sales data may provide incorrect forecasts of future sales, while manual management of inventories may be time-consuming and may also involve errors. Furthermore, ineffective management of obsolete inventory can lead to significant financial losses.

Finally, the technologies discussed, including Microsoft Azure and Power BI, provide the necessary infrastructure and tools to support the implementation of a modern, end-to-end data

pipeline. These technologies enable improved data integration, processing, and visualization capabilities.

In summary, this chapter provides a strong evidence-based foundation for project planning and system design in the following chapters.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted for the development of the proposed Recoverable Assets (RA) and inventory Risk Management (IRM) system. It concentrates on how the system is practically developed and executed in an organized manner, whereas the preceding chapter covered the theoretical underpinnings, related works and technologies involved.

The methodology acts as a guide to guarantee that the development of the process is methodical, structured and in line with the goals of the project. The development of the end-to-end data engineering pipeline in this project is guided by an organized System Development Life Cycle (SDLC) strategy. Every stage of the process is meticulously modified to meet the needs of the analytics, integration and data processing in a club-based setting.

This chapter also will cover the technologies utilized to support the system's implementation, explain each stage of the development process and provide more justification for the methodology that was used. The project guarantees consistency, dependability and efficiency in providing a useful and scalable inventory risk management system by adhering to a clearly defined approach.

3.2 Methodology Choice and Justification

The core methodologies for this research is the Medallion Architecture, a data design pattern that organizes information into a series of functional layers within a data lakehouse environment (Armbust et al., 2021). This architecture utilizes a “multi-hop” process to incrementally improve data quality and structure as it moves through the pipeline (Microsoft, 2024). While alternative frameworks such as the Lambda Architecture –which separates batch and real-time processing paths were considered, they were deemed unsuitable for this project. The Lambda Architecture's dual-path complexity is unnecessary given PPG's batch-oriented data refresh cycle, which does

not require a high-frequency real-time stream processing layer; thus, the Medallion approach was selected for its streamlined maintenance and superior data consistency (Microsoft, 2024).

The primary motivation for this selection is to prevent the creation of a data swamp. A data swamp is an unmanaged data lake that lacks the necessary metadata, governance, and quality controls, rendering the stored information inaccessible or untrustworthy for business decision-making (Gorelik, 2019). By implementing a three-layer structure which are Bronze, Silver, and Gold, the group ensures that data matures from raw ingestion to analytics-ready insights while maintaining full auditability throughout the data engineering lifecycle (Armbrust et al., 2021).

3.2.1 Hierarchical Data Refinement Layers

3.2.1.1 Bronze Layer (The Auditability Argument)

The Bronze layer serves as the landing zone for raw data ingested from the six PPG CSV files, including materials.csv and purchase_orders.csv. Data is stored in its native format to ensure the pipeline remains “non-destructive” (Microsoft, 2024). Since the primary justification for this layer is auditability, immutable record of inventory snapshots is essential for financial accountability as technological obsolescence is a major life cycle driver (Gravier & Swartz, 2009).

This principle of preserving historical data to manage obsolescence risk translates directly into the Bronze layer’s non-destructive ingestion strategy. It allows PPG to perform proactive obsolescence management by providing a historical baseline, ensuring that if the Magna Carta business rules are adjusted, the results can be recalculated from the original source (Gravier & Swartz, 2009). While this layer introduces storage overhead due to data redundancy, it is a necessary trade-off for audit integrity.

3.2.1.2 Silver Layer (The Quality Control Argument)

In the Silver layer, data is cleansed, integrated, and transformed into a validated state (Arnbust et al., 2021). This is where the core PPG business rules, specifically the calculation of Recoverable Assets (RA) and the flagging of Dead Stock, are applied. This layer provides the isolation necessary for incremental quality refinement. According to Arnbust et al. (2021), the Silver layer acts as a cleansing station where data from disparate sources is unified and validated against business schemas.

This is superior to traditional single-step ETL processes because it allows for granular validation (e.g., detecting null values or duplicates in `inventory_snapshot.csv`) to be resolved before complex business logic is applied. However, the complexity of managing these transformations increases the processing latency of the pipeline, which was mitigated by utilizing automated orchestration via Azure Data Factory.

3.2.1.3 Gold Layer (The Performance Argument)

The Gold layer represents the final refined state where data is structured into a star schema for visualization. This structure consists of a central `fact_inventory_status` table surrounded by descriptive dimension tables like `dim_material` and `dim_supplier`. The use of a star schema is justified by the requirement for high query performance. According to the Kimball methodology, dimensional modeling optimizes data retrieval speed for analytical tools like Power BI by minimizing complex table joins (Kimball & Ross, 2013).

This directly supports the project's goal of answering PPGG's seven key business questions such as identifying which sales orders are affected by shortages with minimal response times. While the Gold layer is less flexible for raw data exploration than the Bronze layer, its structured nature ensures that logistics managers receive rapid, consistent, and actionable insights.

3.3 Phases of the Chosen Methodology

3.3.1 Process and Activities within Each SDLC Phase

The SDLC has five major phases in this project, with activities are uniquely adapted to the Recoverable Assets (RA) and Inventory Risk Management (IRM) system development. This method will make sure that the system is created in a systematic and organized way, and every stage will lead to the successful implementation of the data engineering pipeline. The phases are modified to meet the needs of processing and analytics of data in the project.

Phase 1: Planning and Requirement Analysis

The planning and requirement analysis phase establishes the foundation of the project by defining the scope of the problem and identifying system requirements. In this project, the focus is on addressing inventory-related challenges, specifically the determination of excessive inventory, outdated supplies and possible stock-outs.

The important business requirements were based on the requirement to identify Recoverable Asset, Magna Carta Dead Stock, expired materials and the risk of stockout and impact on production and customer orders. To support these objectives, six datasets including materials.csv, inventory_snapshot.csv, production_orders.csv, purchase_orders.csv, suppliers.csv, and sales_orders.csv were analyzed with the help of material_id that is the major common attribute between the datasets.

Moreover, the essential performance metrics like RA classification, MC status, the amount provided, and the stockout risk were formulated. These necessities informed the system design and made sure that the end product meets the intended business purposes.

Phase 2: System Design

The system design phase transforms the identified requirements into a structured

architecture and data model. This stage plays a very significant role in making the system to be scalable, efficient and able to support the operations of the analytical system.

Three-layer data architecture was developed, consisting of bronze, silver, and gold layers. The Bronze layer receives raw data, the Silver layer processes raw data and converts it to data, and finally, the Gold layer stores data ready to be analyzed. Such a layered structure guarantees that there is an unambiguous distinction of data processing steps and enhances maintainability.

Furthermore, the data warehouse was planned to have a Star Schema with the fact table being fact_inventory status and the supporting dimension tables, including dim_material, dim time and dim supplier. This format allows effective querying and building analytical dashboards.

Phase 3: Development (Data Engineering Lifecycle Implementation)

Development is the main part of the project, during which the system is implemented in accordance with the specifications of the design. This stage is subsequent to the Data Engineering Lifecycle to have a formal and scalable data pipeline.

The ingestion of data is done through the Azure Data Factory whereby raw CSV data is loaded into the Azure Data Lake storage. Data consistency is provided by basic data quality checks. This data is then processed in Azure Databricks where it is cleaned, integrated and transformed.

Business logic is used to create important insights during the transformation process. It involves computing 12-month consumption, Recoverable Assets, Dead Stock and expired material, computing provision levels and the stockout risk. The processed data is then stored in a Star Schema in Azure SQL Database.

Finally, the data is visualized in the form of interactive dashboards using Power BI, allowing users to analyze inventory risks and make informed decisions. This step is necessary to make sure that raw data is successfully processed into useful information.

Phase 4: Testing

The testing phase is the part where it involves a series of detailed checks to ensure the technical setup and all of the formulas worked perfectly. Initial activities focused on verifying the accuracy of core calculations, focus for RA classification, expiry detection, and provisioning. Furthermore, The full data pipeline was also tested across the bronze, Silver, and Gold layer to confirm that information moved from the local server to cloud smoothly without any error or missing value. Finally, the result shown on the Power BI dashboard will be compared with the expected record to make sure the visual reports were correct. This phase acted as a final quality check to make sure there was no issue before the system went live.

Phase 5: Deployment and Evaluation

In this phase, the system will be published on the Microsoft Azure platform. The deployment included setting up the data pipeline to run automatically daily on schedule, which removed the need of any manual work that can have a human error. Once the system was running, a full review was conducted to see how well it performed in three main areas which is Accuracy, Efficiency and Scalability. The review confirmed that the data stayed consistent and correct throughout the process, the cloud tools moved the data quickly, and the system was capable of handling more information as the business grows. This final step ensured that the delivery of permanent and automated way to track business structure

3.3.2 Design Modelling

Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how raw CSV data is ingested through Azure Data Factory into Azure Data Lake, processed using Azure Databricks, and stored in Azure SQL Database for analytical reporting in Power BI.



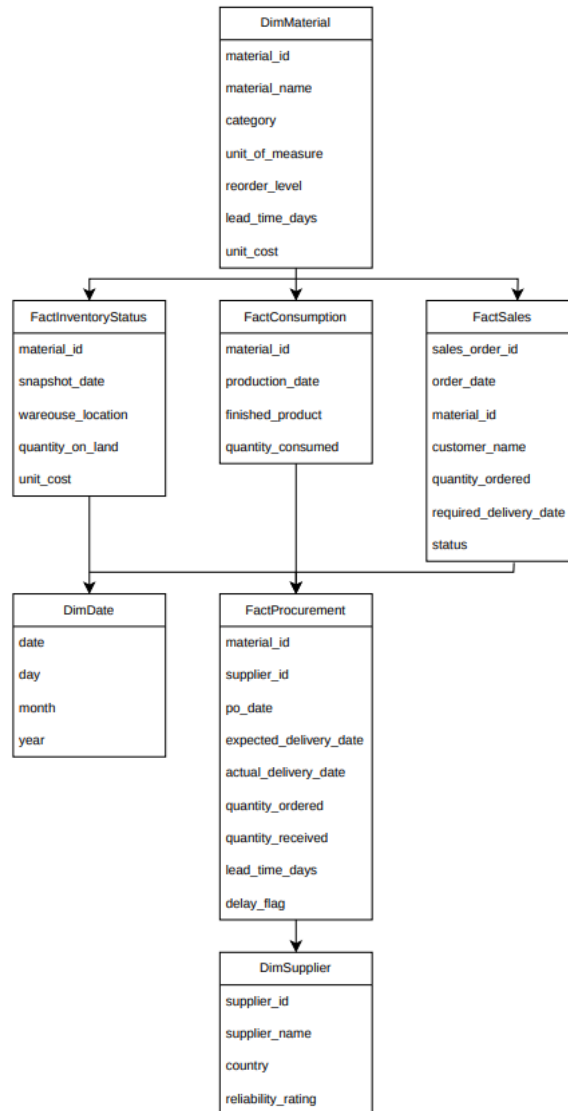
System Architecture Diagram

The system architecture integrates Azure services to support scalable data processing. Azure Data Factory handles data ingestion, Data Lake stores raw and processed data, Databricks performs transformation and analytics, SQL Database stores structured data, and Power BI provides visualization.



Star Schema Diagram

The star schema organizes data into fact tables capturing transactional events and dimension tables providing descriptive context, enabling efficient querying and analysis for inventory risk management.



3.3.3 Design Tools

The design of the Recoverable Assets (RA) and Inventory Risk Management (IRM) system needs clear visualisation of the system architecture, data flow and database structure. To support this, several design tools were used to make sure the system is well organised, easy to understand and follow the data engineering pipeline.

First, [Draw.io](https://draw.io) and Lucidchart were used to create system architecture diagrams and data flow diagrams (DFD). These tools help show how data moves through the three layer architecture, starting from raw data in the Bronze Layer, then processed in the Silver Layer, and finally stored in the Gold Layer for analysis. The diagrams also show how the Azure Data Factory, Data Lake,

Databricks, SQL Database, and Power BI are connected. This makes it easiest to understand how the system works and ensures all components are properly linked.

Next, Power BI Model View was used to design and check the data warehouse schema, especially the star schema in the Gold Layer. It helps define the relationships between the fact table (`fact_inventory_status`) and dimension tables (`dim_material`, `dim_time`, `dim_supplier`). This ensures the data model is correct and suitable for analysis and dashboard use.

Overall, these tools help improve the clarity and understanding of the system design. They also make sure the system is properly structured and meets the project requirements.

3.4 Technology Used Description

This project makes use of multiple Azure tools along with Power BI for designing an end-to-end data engineering pipeline for inventory risk management. The selection of technologies depends on their role in each phase of the data engineering pipeline, such as data ingestion, data storage, data processing and data visualization.

3.4.1 Azure Data Factory

Azure Data Factory is used for managing and automating the data ingestion process. Specifically, in this project, Azure Data Factory is used to extract data from multiple CSV files and load data into the storage layer. The use of Azure Data Factory pipelines allows automating the movement of data and makes the process more reliable and efficient.

3.4.2 Azure Data Lake Storage

Azure Data Lake Storage is used as a main storage layer for all raw data and constitutes a Bronze layer of the data pipeline. The Bronze layer consists of storing data in its initial form before data processing. The reason for using Azure Data Lake Storage is scalability and ability to store large amounts of data. Storing data in the initial form in this layer is helpful for possible future processing of data and allows tracing the origin of data.

3.4.3 Azure Databricks

In this project, Azure Databricks is used for data processing and transformations and performs the most important role in the Silver Layer of the data pipeline. In this phase, cleaning, integrating and transforming the initial raw data take place along with applying business logic to raw data, which includes such components as Recoverable Assets (RA), Dead Stock, expired material and stock out risks. Azure Databricks fits perfectly for this purpose owing to its ability to work with large amounts of data.

3.4.4 Azure SQL Database

The next layer, the Gold Layer, is Azure SQL Database, where the processed data is saved in structured form. Azure SQL database is designed to organize data according to the star schema, involving fact and dimension tables. It is used for conducting analytical processes and provides the best performance.

3.4.5 Microsoft Power BI

The last technology used in this project is Microsoft Power BI. It is used for visualizing the processed data in a form of interactive dashboards. Power BI works directly with Azure SQL database to show the processed data through various types of charts and KPIs.

3.5 System Requirement Analysis

In this section, the system requirements for the Inventory Risk Management system will be discussed. These requirements can be subdivided into functional and non- functional aspects.

3.5.1 Functional Requirements

Here are some functional requirements that the system should meet:

- Data Ingestion

It is necessary to import data from various CSV files, which include such categories of information as material, inventory, production, supplier, purchase order and sales order into the data storage subsystem.

- Data Cleaning and Validation

Data quality control should be performed, including data normalization in terms of dealing with missing data, deleting duplicates and checking data types before processing.

- Data Transformation and Processing

To obtain relevant results, certain actions should be performed by the system such as identifying Recoverable Assets (RA) depending on their consumption, detecting Magna Carta Dead Stock, identifying expired materials and also calculating stockout risks.

- Data Integration

It is required to unify data from different sources into a single dataset that could show the state of inventory at a particular moment.

- Data Storage

To enable data analysis later on, it is needed to organize data using the data warehouse strategy.

- Data Visualization

To provide users with opportunities to analyze data and manage inventory risks, it is necessary to build interactive dashboards.

3.5.2 Non-Functional Requirements

Non-functional requirements are the following:

- Scalability

With growing data, it is important that the performance of the system remains stable.

- Data Quality and Reliability

It is necessary to guarantee data accuracy, consistency and integrity.

- Performance

The system should perform tasks quickly, giving the opportunity to receive results in a timely manner.

- Security

The data storage and analysis should be performed in a protected way preventing any unauthorized actions.

- Usability

Users' dashboards should provide an easy way to analyze data and make conclusions.

- Maintainability

System modification should be easy, especially in case some changes are made to data sources or business logic.

3.6 Chapter Summary

The process for creating the Inventory Risk Management (IRM) and Recoverable Assets (RA) systems has been described in this chapter. The chosen methodology offers a methodical and structured approach to guarantee that the project is produced in an orderly, effective, and scalable way.

The chapter began by outlining how the Medallion Architecture and the System Development Life Cycle (SDLC) approach were appropriate for the suggested data engineering pipeline in order to support the methodology choice. From raw data intake to analytics-ready data for reporting and decision-making, the Bronze, Silver, and Gold levels guarantee appropriate data flow management.

The stages of the development process, such as planning and requirement analysis, system design, development, testing, and deployment with assessment, were also thoroughly covered. Each stage was especially modified to meet the requirements of cloud-based data processing and inventory risk management.

The design modeling tools and system requirements that facilitate the system's successful implementation were also covered in this chapter. The technologies that were chosen, such as Power BI and Microsoft Azure services, were also described according to their functions in data ingestion, storage, transformation, and visualization.

Overall, this chapter's methodology provides a solid basis for the project's subsequent phase, which concentrates on system analysis, design, and implementation specifics. The methodical methodology guarantees that the suggested solution may successfully accomplish the project's goals while promoting future scalability and maintainability.

CHAPTER 4: Analysis and Design

4.1 Introduction

This chapter is about the system analysis and design phases for the Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The system analysis phase evaluates current operations to identify core business challenges and create the data collection plan required to build the data data pipeline. Based on these findings, the technical scope is outlined through detailed functional and non-functional requirements. Together, these elements form a blueprint that will be used to develop the multi layer azure pipeline and follow with power BI dashboard

4.2 System Analysis

4.2.1 Case Study

This project is grounded in a case study of the manufacturing industry, with PPG Industries, a global company specializing in paints and coatings and specialty materials, as the subject. PPG Industries has large-scale manufacturing and supply chain operations that demand effective inventory management to ensure smooth production and material availability (PPG Industries, 2024). PPG Coatings (Malaysia) Sdn Bhd provides support in the procurement, supply chain management, inventory operations and shared service functions of manufacturing and operation in the Asia Pacific region (Malaysian Investment Development Authority [MIDA], 2023).

Proper management of inventory in various operational systems and departments is one of the significant problems in manufacturing industries. Often, inventory information like stock levels, supplier deliveries, production usage, expiry dates, etc., and customer demand is maintained separately, giving little real-time and centralized visibility of the inventory situation. This disjointed world has a negative impact on the visibility of inventory and slows down the decision-making process.

The project specification identifies several inventory risks that should be monitored, such as excess inventory, dead stock, expired materials, and stockout risks. Materials older than 12

months of use can be declared Recoverable Assets (RA), and materials that have not been used for a long time can be classified as Magna Carta Dead Stock. Furthermore, low inventories can have an impact on production runs and the number of sales orders placed by the customers.

The existing inventory monitoring is also primarily done manually and through periodic reporting, which is time-consuming and may involve human error. Gravier and Swartz (2009) suggest that a lack of good inventory control and obsolescence management can have a large cost and affect supply chain performance. Late identification of inventory problems could result in losses of money, poor use of warehouse space, production delays and losses in operational efficiency.

The system that is proposed is a centralized inventory risk management system with an end-to-end data engineering pipeline to overcome these challenges. The system connects various datasets related to inventory into a unified analytics context to enhance inventory visibility and system monitoring. The proposed solution will allow the automatic identification of Recoverable Assets (RA), Dead Stock, Expired Materials, Stockout Ranks, and Affected Customer Sales Orders. In addition, interactive dashboards and analytical reports enable quicker and better decision-making for inventory management operations.

In conclusion, the case study exemplifies how centralized analytics and automated inventory monitoring systems can help the manufacturing industry enhance operational efficiency and minimize risks associated with inventory.

4.2.2 Company Organization Structure

PPG Industries operates using a function-based organizational structure that supports manufacturing, procurement, supply chain management, finance, and information technology operations. These departments support manufacturing activities, inventory monitoring, and operational decision-making (PPG Industries, 2024).

In Malaysia, PPG Coatings (Malaysia) Sdn Bhd serves as one of PPG's regional operational and shared service centres supporting Asia Pacific operations. According to the Malaysian Investment Development Authority (MIDA, 2024), the shared service centre manages

functions such as finance, procurement, supply chain, information technology (IT), and master data management. These departments are directly related to the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system.

Firstly, the Supply Chain and Procurement department is responsible for coordinating with suppliers, carrying out procurement, and replenishing inventories. According to PPG Global Procurement (2024), PPG's procurement operations manage raw material sourcing and supplier relationships to support manufacturing continuity. This department is crucial in the proposed system since its performance can directly impact the levels of stock and production operations.

Next, the Warehouse and Inventory Management team is responsible for inventory storage, stock control, and material handling. This department is directly in charge of inventory monitoring, including excess inventory, expired materials, and inventory risk of stock out for the project case study. Good inventory management is crucial for efficient warehouse operations and minimising inventory risks.

In addition, the Production department is responsible for manufacturing and production scheduling activities. Adequate availability of raw material is critical to production operations as it can cause production disruption. Hence, it is required to maintain a raw materials inventory system continuously to achieve a smooth production process and to avoid a shortage of materials.

Meanwhile, the Finance department shall keep track of inventory-related financial exposure and inventory write-down risks. Finance activities are one of the shared service functions of the PPG Malaysia region, according to MIDA (2024). The importance of this department is that excess inventory and obsolete materials can have a direct impact on the costs and financial reporting of this organization.

Furthermore, the Information Technology (IT) department provides enterprise systems, databases, and digital operations for the organisation. MIDA (2024) reported that IT is one of the key operational services in PPG Malaysia's shared services centre. The role of the IT department in this project is crucial, as the inventory risk management system will be reliant on cloud-based platforms, databases, and business intelligence tools like Microsoft Azure and Power BI.

In general, the integration of procurement, warehouse, production, finance, and IT is crucial for successful inventory management and decision-making in the Company.

4.2.3 System Requirements Gathering Techniques

Several requirement gathering techniques were used to identify the business and system requirements for the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system.

The first step was a literature review on concepts concerning Inventory Management, Data Engineering Pipelines, ETL/ELT processes, Cloud Computing and Business Intelligence systems. Academic journals and the documentation of Microsoft Azure were reviewed, along with previous research, to find appropriate technologies and approaches for the proposed system. This method has been used to provide the theoretical and technical underpinning mentioned in Chapter 2.

Next, the case study analysis was conducted according to the specifications of the industry project, PPG. The case study furnished information on inventory challenges, including overstocking, out-of-stock, expiring materials, dead stock, and so on. The project specification also outlined the data required, business rules, and analytical questions that were necessary for the process of system development. Based on this analysis the functional requirements and reporting requirements of the proposed system are identified.

Furthermore, document analysis was performed based on the descriptions of documents provided in the dataset, the requirements of the project, and the information about the business

processes. The analysis was centred on the understanding of how the data is related, what the attributes of the inventory are, and what operational requirements are pertinent to inventory monitoring and analytics. This approach contributed to the data pipeline development, database design, and dashboard needs.

In addition, interviews and discussions were conducted during the industrial visit to PPG Industries to better understand the operational workflow and inventory-related challenges associated with the case study. The project team also created a set of questions before the visit related to the case study. The session consisted of an explanation by the person in charge of PPG regarding the background of PPG, goals of the project, the datasets and business rules to be used, and the analytical outputs that were expected for the proposed system. Group meetings for discussion and clarification were also held throughout the project with the lecturer to resolve misunderstandings, further develop the system requirements, and confirm that the proposed solution was in line with the project goals.

Overall, the combination of literature review, case study analysis, document analysis, and interviews ensured that the proposed system requirements were practical, relevant, and in line with the industrial case study.

4.3 System Requirements

4.3.1 Functional Requirements

Requirements Category	Description
Raw Data Ingestion (Bronze)	ingest 6 operational source CSV (materials.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, suppliers.csv, and sales_orders.csv) into a secure cloud data lake while keep their original raw formats.
Data Quality Guardrails (Silver)	Automatically validate data type,handle null values, and remove duplicate records
Recoverable Assets (RA)	Calculate 12 month based on the consumption rate and flag any inventory stock volume that exceeds this threshold as a Recoverable Asset (RA)
Magna Carta Dead Stock Flagging	Flag unexpired materials with more than equal 9 consecutive months of zero operational consumption
Magna Carta Expired Flagging	Compare active calendar timelines against raw lot parameters to flag it as Magna Carta Expired upon passing their designated lot expiry date
Financial Provisioning	Calculate a 100% financial write down provision for all flagged MC Dead and Expired Stock
Stockout Risk & Traceability	Flag materials running low relative supplier lead times, identify unfulfilled production orders, and trace the downstream impact to flag affected open customer sales orders and their associated clients
Star Schema Serving (Gold)	Structure processed attributes into a star schema, attaching all metrics around a central fact table.
Interactive Dashboards	Reporting interface must display final calculations by using a Power BI dashboard to solve the 7 specific business question regarding inventory positions and risk metrics.

4.3.2 Non Functional Requirements

Requirements	Performance Standard
Data Integrity	Logic transformations used within cloud pipeline must maintain absolute data integrity, ensuring summaries perfectly match original source values
Query Performance	The dashboards Power BI must be optimized to render visual tiles and respond filtering queries within the acceptable operational limit
Scalability	The Azure infrastructure must capable to handling increasing volume of transactional snapshot rows over time without lowering the performance
Maintainability	Pipeline workflow within Microsoft Azure must follow structured design allowing engineering teams to easily update business logic formulas
Security and Access	Limit access to data lake, warehouse and reports dashboard to authenticated, authorized users

4.4 Current System Analysis

Most manufacturing companies including the PPG industry case study use traditional and semi-manual inventory monitoring systems. These are typically Excel spreadsheets, simple ERP reporting applications and periodic manual reporting from various departments like procurement, warehouse and finance.

Typically, inventory data is not fully integrated in the current environment, and exists in isolated systems. For instance, production data is kept by the production team, sales data is handled by the sales team and inventory is controlled by the warehouse team. This results in a disjointed data model, with data not being updated in real time, and that is hard to integrate.

This has left managers with a lack of timely information to make decisions, as they have to manually integrate data from various sources to identify inventory risks, including excess stock, dead stock or stockout conditions. This raises the possibility of human mistakes, inconsistent data and out-of-date reporting.

Another significant constraint of the existing system is the absence of predictive or automated inventory risk classification. For example, items used more than 12 months ago or items that have gone “out of stock” are not automatically marked as such and could only be discovered after losses have occurred.

Besides, it is very manual and time consuming to identify the impact of stockouts on production orders and customer sales orders. This lowers the efficiency of the operation and makes it hard for companies to react swiftly to disruptions in the supply chain.

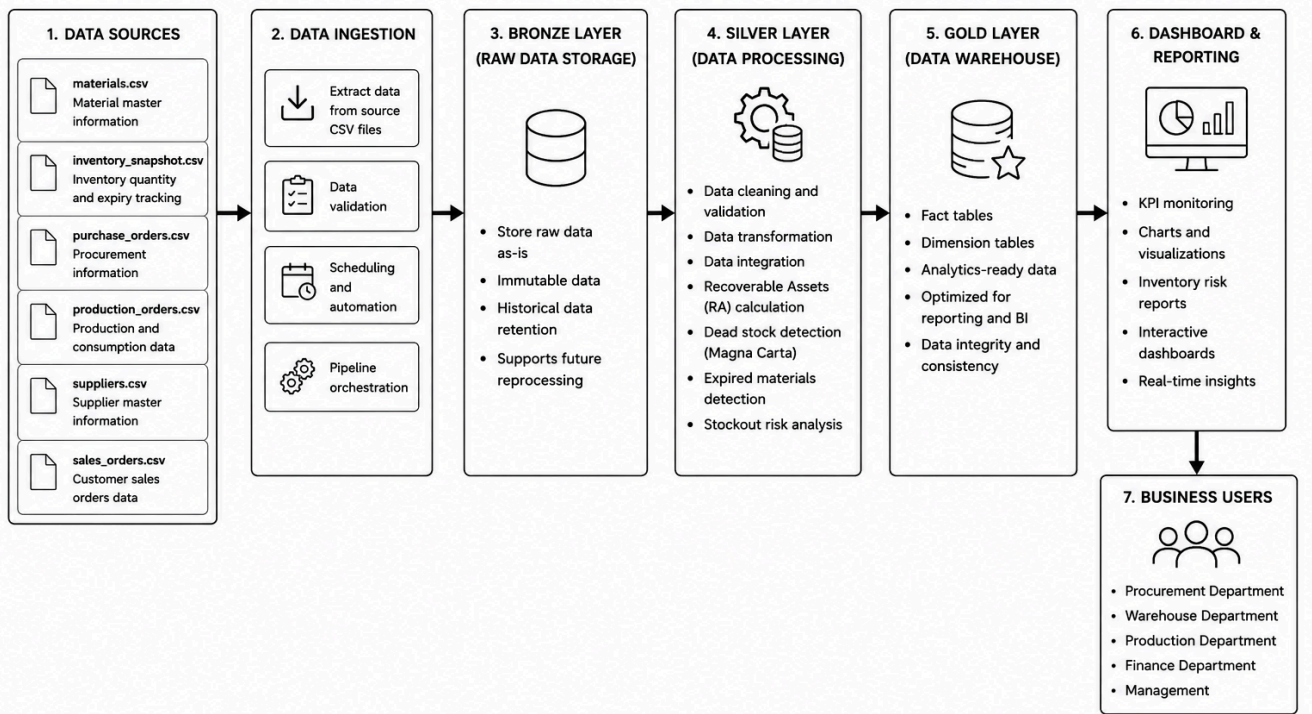
So, there is a need for an enhanced automated data engineering pipeline. The proposed system with Microsoft Azure and Power BI will address these problems by consolidating all data sources within a single platform, adding automated business logic and offering real-time analytics dashboards for improved decision making.

4.5 System Design

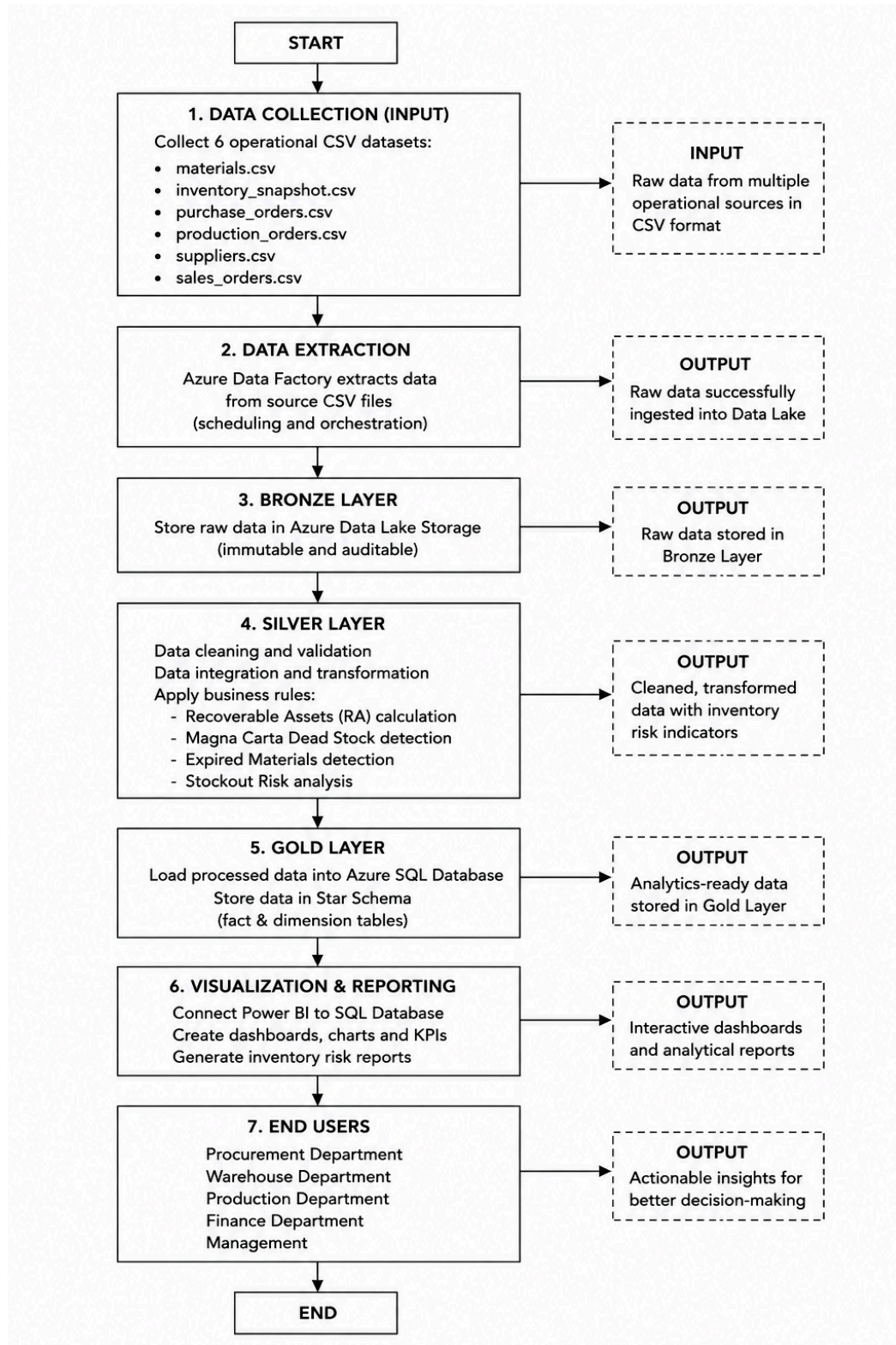
4.5.1 System Architecture

Microsoft Azure services and Microsoft Power BI are used in the cloud-based architecture of the suggested Recoverable Assets (RA) and Inventory Risk Management (IRM) solution. To provide effective data management, scalability, and data quality, the architecture employs the Bronze, Silver, and Gold layers of the Medallion Architecture. Inventory risk monitoring and decision-making are supported by the system's ability to convert unprocessed inventory data into valuable business insights.

Data Sources, Azure Data Factory, Azure Data Lake Storage, Azure Databricks, Azure SQL Database, and Power BI Dashboard are the six primary parts of the overall architecture. Each layer in the data engineering pipeline carries out a specific task as data moves successively through these components.



System Flow



System Modules

Modules	Functions
Data Ingestion Module	Extracts data from source CSV files and loads them into Azure Data Lake Storage
Data Storage Module	Stores raw inventory data in Bronze Layer
Data Processing Module	Cleans, transforms and validates inventory data
Inventory Risk Analysis Module	Identifies Recoverable Assets, Dead Stock, Expired Materials and Stockout Risks
Data Warehouse Module	Stores analytics-ready data in Star Schema format
Visualization	Displays KPIs, reports and dashboards through Power BI

System Inputs

The proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) System receives data from six operational datasets provided by PPG Industries. These datasets are used to analyze inventory conditions, identify risks, and support decision-making processes.

The system receives data from six datasets:

- materials.csv
- inventory_snapshot.csv
- purchase_orders.csv
- production_orders.csv
- suppliers.csv
- sales_orders.csv

System Outputs

The system produces analytical reports and dashboards to support inventory risk management and business decision-making.

- The system generates:
- Recoverable Assets Report
- Magna Carta Dead Stock Report
- Expired Materials Report
- Stockout Risk Report
- Financial Provision Report
- Affected Customer Sales Orders Report
- Interactive Power BI Dashboards

4.5.2 Explanation of System Architecture Components

1. Data Collection (Input)

This component acts as the system's entry point. Six operational datasets, materials, inventory snapshots, purchase orders, production orders, supplier records, and sales orders are used to gather inventory-related data. The raw data needed for inventory risk analysis is provided by these datasets.

2. Data Extraction

Azure Data Factory (ADF) is responsible for extracting data from the source CSV files. It automates the data ingestion process and ensures that data is collected consistently and efficiently. ADF also manages scheduling and workflow orchestration for the entire data pipeline.

3. Bronze Layer

The Bronze Layer stores raw data in Azure Data Lake Storage (ADLS). Data is maintained in its original format without any modifications. This layer acts as a centralized repository for historical records and supports data auditing and traceability.

4. Silver Layer

Azure Databricks functions as the data processing and transformation engine. In this layer, data is cleaned, validated, standardized, and integrated across different datasets. Business logic is also applied to identify inventory risks, including:

- Recoverable Assets (RA)
- Magna Carta Dead Stock
- Magna Carta Expired Materials
- Stockout Risks
- Financial Provision Calculations

The Silver Layer ensures that data is accurate, consistent, and ready for analytical processing.

5. Gold Layer

Azure SQL Database functions as the centralized data warehouse and the Gold Layer. Fact and dimension tables make up the Star Schema structure used to arrange processed data from Databricks. In addition to enabling effective reporting and dashboard creation, this approach enhances query performance.

6. Visualization and Reporting

Microsoft Power BI is the visualization and business intelligence platform used in the proposed system. It retrieves processed data from Azure SQL Database and presents information through interactive dashboards, charts, KPIs, and reports. Power BI allows users to monitor inventory conditions and identify risks in real time.

7. End Users

The final users of the system include Procurement Officers, Warehouse Managers, Production Managers, Finance Analysts, and Senior Management. These users utilize dashboard insights to make informed decisions regarding inventory control, procurement planning, financial provisioning, and operational risk management.

4.5.3 System Design

The PPG inventory risk management pipeline consists of the physical and logical designs that are needed to help convert raw operational data into meaningful business intelligence. Chapter 9, as well as ELT, EL or ETL, talked about both the ETL, EL and ELT system in an abstract way, but this system is a formalized ELT (Extract, Load, Transform) system. With this architecture, Azure Data Factory can quickly load raw data into a scalable lake storage, and then perform complex

transformations that use more compute later in the pipeline with Azure Databricks (Armbrust et al., 2021).

Data Engineering Process

The data engineering process is an automated, multi-hop pipeline that incrementally evolves data maturity by the Medallion Architecture (Armbrust et al., 2021). The first step is to load the static source files (the Bronze layer) into a data lake. By keeping this ingestion part of the pipeline separate from the transformation part, the environment will not turn into an unmanaged 'data swamp', with no metadata or governance (Gorelik, 2019) .

After being secured, data is transferred to the Silver layer for comprehensive data cleansing, type validates, and integration. It is at this point that the complicated PPG business logic is programmed – for example, the 12-month consumption limits are calculated and the Magna Carta thresholds are determined. Finally, the processed data is aggregated and organized in a dimensional star schema in Gold, for optimal performance of queries on the data.

Data Source

The pipeline uses synthetic operational information that is generated to mimic PPG's real-world supply chain. This data is available in six different Comma-Separated Values (CSV) files. In this schema, `material_id` is used as the primary relational link, it is the same key field and can be used in most of the data sets to track the material from supplier to customer from a comprehensive perspective:

- **materials.csv**: This is the base reference attributes table for materials, by category (Pigment, Resin, Solvent, Packaging) and is the primary dimension source.
- **inventory_snapshot.csv**: Records the stock levels and lot expiry dates every month. In this file, there is a significant amount of information that is essential for answering business question Q4 on Magna Carta Expired classifications.
- **purchase_orders.csv**: contains Inbound Transactions and late deliveries by supplier, which is needed for analyzing procurement efficiency.
- **suppliers.csv**: Stores the master information of the vendors such as geographic locations, reliability etc., which is closely associated with purchase orders data.

- **production_orders.csv**: This is a file that shows the production orders, and it is used to see if there are any unfulfilled production orders (Q6).
- **sales_orders.csv**: This cross-linking can be done with the inventory data using material_id so that the impact of the material stockout on "affected" customers sales can be traced (Q7).

Extraction, Transformation, and Loading (ETL)

To successfully execute the pipeline, it uses the coordination of certain Azure resources, coordinating the Extraction, Transformation, and Loading (ETL) steps to ensure they are executed correctly:

- **Extraction (Ingestion to Bronze)**: Azure Data Factory (ADF) is responsible for orchestrating the phase. The "Copy Data" activities are used to get the six CSV files that are the source data and store them as immutable in Azure Data Lake Storage (ADLS Gen2). This is the Bronze layer, which meets the need for a historical audit trail, with data being ingested without any loss of data and without any destruction (Gravier & Swartz, 2009).
- **Transformation (Refinement to Silver)**: Azure Databricks is used to perform the transformation phase (Refinement to Silver). Using PySpark and SQL notebooks that are connected to scalable compute clusters, data engineers clean the raw data. This stage deals with nulls, formatting dates to be consistent for lot expiry and the merging of the various CSV files with material_id. Computation of core business rules here, which creates definitive Recoverable Assets (RA) and Magna Carta Dead Stock flags.
- **Loading (Serving to Gold)**: This is the last step to validate and transform the data from Databricks and push to Azure SQL Database. The data is loaded into a star schema design that is pre-defined (Kimball & Ross, 2013). This schema has centralized Fact tables, that are surrounded by descriptive dimension tables like DimDate, DimMaterial and DimSupplier, namely FactInventoryStatus, FactConsumption, FactSales and FactProcurement.

- **Visualization (Microsoft Power BI accesses optimized SQL Database):** Since the ELT pipeline runs on a scheduled batch-refresh cycle, as opposed to streaming real-time data, Import mode was chosen. This mode stores the data into memory, allowing for better visualization performance, and makes sure that the dashboard can quickly respond to logistics managers' unique business questions without incurring query latency (Microsoft, 2023).

4.5.4 Database Design

In the initial phase of database modeling, this project adopts a Non-SQL / Document-Oriented Data Modeling approach to structure the raw data ingested into the Bronze Layer of Azure Data Lake Storage. This non-relational (schemaless) method was selected so that the six source files belonging to the organization can be stored in their native formats instantly. This preserves historical data integrity through a non-destructive ingestion strategy.

In this Non-SQL model, data is not forced into rigid table rows and columns. Instead, it is mapped as independent collections of documents, where logical connections between different documents are flexibly established using the common key attribute `material_id`. The following outlines are designed for each data collection based on the actual attributes:

a. `materials.csv`

Attribute	Data Type	Description
<code>material_id</code>	String	Unique identifier for the material
<code>material_name</code>	String	Name of the material
<code>category</code>	String	Material classification
<code>unit_of_measure</code>	String	Measurement unit
<code>reorder_level</code>	Number	Minimum stock to reorder
<code>lead_time_delays</code>	Number	Supplier delivery lead time in days

unit_cost	Number	Cost per unit of material
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b. inventory_snapshot.csv

Attribute	Data Type	Description
snapshot_id	String	Unique identifier for snapshot log
material_id	String	Reference key to the materials dataset
snapshot_date	Date	Date when the inventory snapshot was recorded
warehouse_location	Number	Physical storage location
quantity_on_hand	Number	Total physical stock available in the warehouse
expiry_date	Date	Material expiration date

c. Production_orders.csv

Attribute	Data Type	Description
production_order_id	String	Unique identifier for the production run
material_id	String	Reference key to the materials dataset
production_date	Date	Date of the material consumption

finished_product	String	Name of the resulting output product
quantity_consumed	Number	Quantity of material used

d. Sales_orders.csv

Attribute	Data Type	Description
sales_order_id	String	Unique identifier for the customer sales order
order_date	Date	Date when the customer places the order
material_id	String	Reference key to the materials dataset
customer_name	String	Name of the client or distributor
quantity_ordered	Number	Volume of product requested by customer

e. Purchase_orders.csv

Attribute	Data Type	Description
po_id	String	Unique identifier for the procurement order
material_id	String	Reference key to the materials dataset
supplier_id	String	Reference key to the

		suppliers dataset
po_date	Date	Date the purchase order was issued
expected_delivery_date	Date	Estimated arrival date promised by the vendor

f. Suppliers

Attribute	Data Type	Description
supplier_id	String	Unique identifier for the vendor
supplier_name	String	Name of the supplier
country	String	Country of origin for the vendor
reliability_rating	Number	Performance rating score of the vendor

4.5.5 Interface Design

This section presents the interface design of the Recoverable Assets (RA) and Inventory Risk Management (IRM) dashboard. The interface acts as the final consumption layer of the data engineering pipeline, transforming the curated star schema in the Gold Layer into actionable visuals for procurement, warehouse, production, finance, and management users. The dashboard is developed using Microsoft Power BI Desktop, which is chosen for its native integration with Azure SQL Database and its rich library of interactive visuals.

The dashboard is developed through a four-step process to ensure that the final interface reflects the project's business requirements:

- Step 1 - Requirement Mapping: Each business question (Q1 to Q7) is mapped to a specific visual need so that every chart on the dashboard serves a clear analytical purpose.
- Step 2 - Wireframing: Low-fidelity wireframes are produced using Draw.io to define the dashboard layout, KPI placement, and navigation flow before any Power BI development begins.
- Step 3 - Data Connection and Modelling: Power BI Desktop is connected to the Azure SQL Database in Import mode, and the star schema relationships between the fact and dimension tables are validated in the Model View. DAX measures are then created for KPIs such as Total MC Provision, RA Count, and Stockout Risk Count.
- Step 4 - Visual Development and Publishing: The visuals are built on the canvas, slicers and drill-through actions are added for interactivity, and the dashboard is tested against the seven business questions before being published to the Power BI Service for end-user access.

The dashboard is organised into four pages to avoid visual overload and to align each page with a specific user role, as summarised in Table 4.6.1.

Page	Title	Purpose and Business Questions Addressed
Page 1	Executive Overview	KPI cards summarising total MC provision (Q5) and counts of RA, MC Dead Stock, MC Expired, and stockout-risk materials. For senior management.
Page 2	Inventory & Reorder Status	Materials below reorder level (Q1) and Recoverable Assets (Q2). For warehouse and procurement teams.
Page 3	Magna Carta Risk Analysis	MC Dead Stock (Q3), MC Expired (Q4), and total financial write-down (Q5). For finance analysts.

Page 4	Production & Customer Impact	Unfulfilled production orders (Q6) and affected open sales orders (Q7). For production planners and account managers.
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Table 4.6.1: Dashboard page structure and alignment with the seven business questions.

Chart Selection and Justification

The choice of chart type follows the principle that the visual representation must match the nature of the underlying data and the question being asked. Table 4.6.2 summarises the mapping between business questions and chart types.

ID	Business Question	Chart Type	Justification
Q1	Materials below reorder level	Table with conditional formatting + bar chart	Table shows exact stock vs reorder level; bars highlight severity of the gap.
Q2	Materials classified as Recoverable Assets	Stacked column chart by category	Compares RA volume across material categories.
Q3	Materials qualifying as MC Dead Stock	KPI card + drill-through table	KPI for headline count; drill-through reveals the material list on demand.
Q4	Materials qualifying as MC Expired	KPI card + timeline chart	Timeline shows when materials expire, supporting proactive write-off planning.
Q5	Total Magna Carta provision required	Large KPI card	A single monetary figure is best shown as a prominent KPI.

Q6	Unfulfilled production orders	Matrix visual + line chart	Matrix shows order-level detail; line chart highlights trends over time.
Q7	Open sales orders and customers affected by shortage	Drill-through table from material to customer	Traces the end-to-end flow from material shortage to impacted customer order.

Table 4.6.2: Mapping of business questions to chart types.

4.6 Chapter Summary

This chapter discussed the system analysis and design of the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The analysis began by examining the current inventory management practices and identifying several challenges, including fragmented data sources, manual monitoring processes, limited real-time visibility, and difficulties in identifying inventory risks efficiently. Based on these findings, the functional and non-functional requirements of the proposed system were established.

In addition, the chapter presented the overall system architecture and design of the solution. The proposed system adopts the Medallion Architecture, which consists of the Bronze, Silver, and Gold layers to support data ingestion, transformation, storage, and analytics. Several Microsoft Azure services, including Azure Data Factory, Azure Data Lake Storage, Azure SQL Database, and Power BI, were selected to build an integrated data engineering and business intelligence platform. The responsibilities of each layer and component were also explained in detail.

Furthermore, the chapter described the system workflow, database design, data processing modules, and dashboard interface design. The proposed solution is capable of identifying Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired Materials, stockout risks, financial provisions, and customer order impacts through automated data processing and reporting. The dashboard design was aligned with the identified business requirements to ensure that users can easily monitor inventory conditions and make informed decisions.

Overall, the proposed system design provides a structured, scalable, and efficient approach to inventory risk management. The design serves as the foundation for the implementation phase, which will be discussed in the following chapter.

Chapter 5 : Implementation and Testing

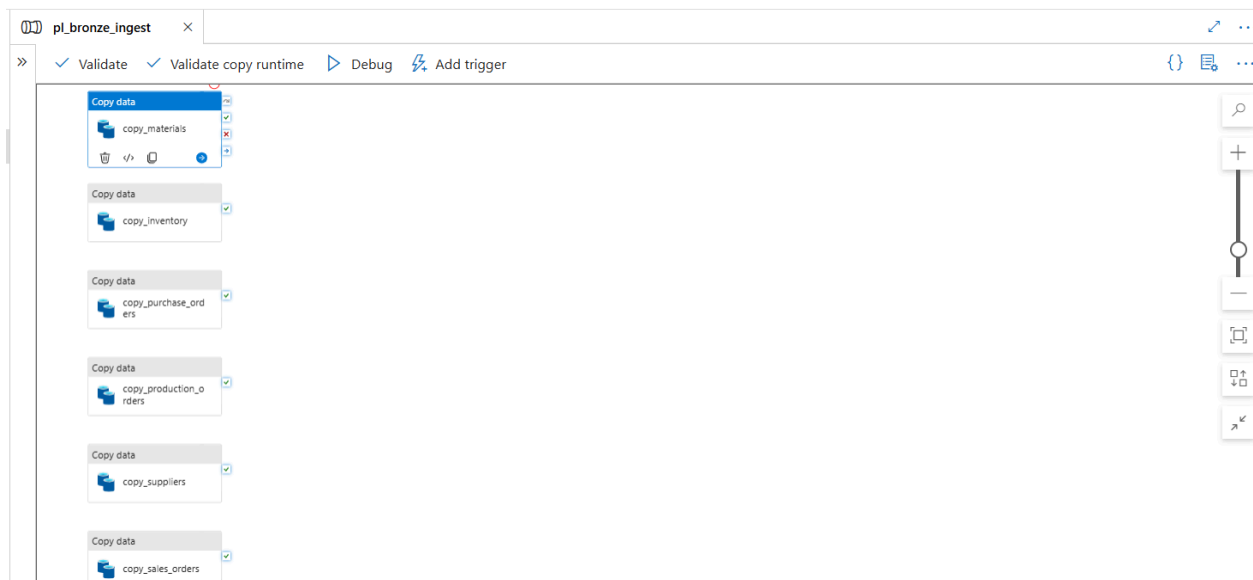
5.1 Introduction

This Chapter focuses on the implementation and testing of the cloud based inventory analysis system developed for the PPG industry. The implementation phase is focused on the use of Microsoft Azure service and Medallion Architecture framework to make the system design happen. The system ingest multiple operational dataset, processes them through different data layers and generates meaningful information via interactive dashboard. Furthermore, In order to make sure the system reaches the expected goals, The test activities which is Black Box tests are being conducted to evaluate the functionality and availability of the system. The result of this test will verify whether the develop system is running as expected and meet the requirements of the project.

5.2 Coding of System Main Functions

5.2.1 Bronze Layer - Azure Setup & Ingestion

Azure Data Factory



This picture shows the Azure Data Factory pipeline named pl_bronze_ingest. The pipeline contains six Copy Data activities, one for each CSV file (materials, inventory, purchase_orders, production_orders, suppliers, and sales_orders). Each activity copies one CSV file from the

source location into the Bronze container in Azure Data Lake Storage. This pipeline is the main entry point of the system and is used to load all raw data into the cloud in one run.

Audit Query 1 - materials

```

1  SELECT
2      COUNT(*)                                AS total_rows,
3      COUNT(DISTINCT material_id)              AS distinct_material_ids,
4      SUM(CASE WHEN material_id IS NULL THEN 1 ELSE 0 END) AS null_material_ids,
5      SUM(CASE WHEN material_name IS NULL THEN 1 ELSE 0 END) AS null_material_names,
6      SUM(CASE WHEN unit_cost IS NULL THEN 1 ELSE 0 END) AS null_unit_costs,
7      SUM(CASE WHEN reorder_level IS NULL THEN 1 ELSE 0 END) AS null_reorder_levels,
8      COUNT(DISTINCT category)                AS distinct_categories
9  FROM OPENROWSET(
10     BULK 'https://sappgg4.dfs.core.windows.net/bronze/materials/materials.csv',
11     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
12 ) AS r;

```

This picture shows the SQL audit query for the materials.csv file. The query uses OPENROWSET to read the file directly from the Bronze container in the data lake. It counts the total rows, checks for null values in important columns such as material_id, material_name, unit_cost, and reorder_level, and counts the distinct categories. The result confirms that 20 materials were loaded with no missing values and 4 distinct categories, which proves that the ingestion was successful.

Audit Query 2 - inventory_snapshot

The screenshot shows a SQL IDE interface with a query editor, a results pane, and a properties pane. The query editor contains the following SQL code:

```

1  SELECT
2      COUNT(*)                                AS total_rows,
3      COUNT(DISTINCT material_id)              AS distinct_material_ids,
4      COUNT(DISTINCT snapshot_date)            AS distinct_snapshot_months,
5      SUM(CASE WHEN quantity_on_hand IS NULL THEN 1 ELSE 0 END) AS null_quantity,
6      SUM(CASE WHEN quantity_on_hand < 0 THEN 1 ELSE 0 END) AS negative_quantity_rows,
7      COUNT(DISTINCT warehouse_location)       AS distinct_warehouses
8  FROM OPENROWSET(
9      BULK 'https://sappgg4.dfs.core.windows.net/bronze/inventory/inventory_snapshot.csv',
10     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
11 ) AS r;

```

The results pane shows the following data:

total_rows	distinct_material_ids	distinct_snapshot_months	null_quantity	negative_quantity	distinct_warehouses
240	20	12	0	0	5

The properties pane on the right shows the query's name as "Audit Query 2 - inventory_snapshot" and its type as ".sql script".

This picture shows the audit query for the inventory_snapshot.csv file. The query checks the total number of rows, the number of distinct snapshot months, and any null values in the key columns.

The result shows 240 rows across 12 distinct months with no nulls, which confirms that the monthly inventory data was loaded completely and is ready to be used in the Silver and Gold layers for stock-level analysis.

Audit Query 3 - purchase_orders

```
1 SELECT
2     COUNT(*)                                AS total_rows,
3     COUNT(DISTINCT po_id)                   AS distinct_po_ids,
4     SUM(CASE WHEN actual_delivery_date = '' THEN 1 ELSE 0 END) AS open_undelivered_pos,
5     SUM(CASE WHEN CAST(quantity_received AS INT) > CAST(quantity_ordered AS INT)
6           THEN 1 ELSE 0 END)                 AS over_delivered_rows,
7     SUM(CASE WHEN CAST(quantity_received AS INT) < CAST(quantity_ordered AS INT)
8           AND actual_delivery_date != ''
9           THEN 1 ELSE 0 END)                 AS partial_delivery_rows
10  FROM OPENROWSET(
11      BULK 'https://sappgg4.dfs.core.windows.net/bronze/purchase_order/purchase_orders.csv',
12      FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
13  ) AS r;
```

This picture shows the audit query for the purchase_orders.csv file. The query counts the total rows, the distinct purchase order IDs, and identifies open undelivered purchase orders, over-delivered rows, and partial deliveries. The result confirms that 40 purchase orders were ingested with one open undelivered PO (PO-002). This query helps verify that procurement data is complete and that special delivery situations can be detected.

Audit Query 4 - production_orders

```
1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT production_order_id) AS distinct_production_orders,
4     COUNT(DISTINCT material_id) AS distinct_materials_consumed,
5     COUNT(DISTINCT finished_product) AS distinct_finished_products,
6     SUM(CASE WHEN quantity_consumed IS NULL THEN 1 ELSE 0 END) AS null_quantity_consumed,
7     MIN(production_date) AS earliest_production_date,
8     MAX(production_date) AS latest_production_date
9 FROM OPENROWSET(
10     BULK 'https://sappgg4.dfs.core.windows.net/bronze/production_orders/production_orders.csv',
11     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
12 ) AS r;
```

total_rows	distinct_production_orders	distinct_materi...	distinct_finishe...	null_quantity_c...	earliest_produ...	latest_producti...
336	84	15	10	0	2025-01-15	2025-12-05

This picture shows the audit query for the production_orders.csv file. The query counts the total rows, the distinct production order IDs, distinct materials consumed, distinct finished products, and the earliest and latest production dates. The result confirms 336 rows, 10 finished products, and zero null values in the quantity_consumed column. This proves that the production data is complete and ready to be used for the Q6 unfulfilled production order analysis.

Audit Query 5 - suppliers

```
1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT supplier_id) AS distinct_supplier_ids,
4     SUM(CASE WHEN supplier_id IS NULL THEN 1 ELSE 0 END) AS null_supplier_ids,
5     COUNT(DISTINCT country) AS distinct_countries,
6     COUNT(DISTINCT reliability_rating) AS distinct_ratings
7 FROM OPENROWSET(
8     BULK 'https://sappgg4.dfs.core.windows.net/bronze/suppliers/suppliers.csv',
9     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
10 ) AS r;
```

total_rows	distinct_supplier_ids	null_supplier_ids	distinct_countries	distinct_ratings
10	7	0	3	3

This picture shows the audit query for the suppliers.csv file. The query counts the total rows, distinct supplier IDs, distinct countries, and distinct reliability ratings. The result shows 10 suppliers from 7 countries with 3 different reliability ratings and no null supplier IDs. This confirms that the supplier reference data is complete, which is important because suppliers are linked to purchase orders for analysing procurement performance.

Audit Query 6 - sales_orders

```

1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT sales_order_id) AS distinct_order_ids,
4     SUM(CASE WHEN status = 'Open' THEN 1 ELSE 0 END) AS open_orders,
5     SUM(CASE WHEN status = 'Delivered' THEN 1 ELSE 0 END) AS delivered_orders,
6     COUNT(DISTINCT customer_name) AS distinct_customers,
7     COUNT(DISTINCT finished_product) AS distinct_products,
8     MIN(required_delivery_date) AS earliest_delivery,
9     MAX(required_delivery_date) AS latest_delivery
10 FROM OPENROWSET(
11     BULK 'https://sappgg4.dfs.core.windows.net/bronze/sales\_order/sales\_orders.csv',
12     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
13 ) AS r;

```

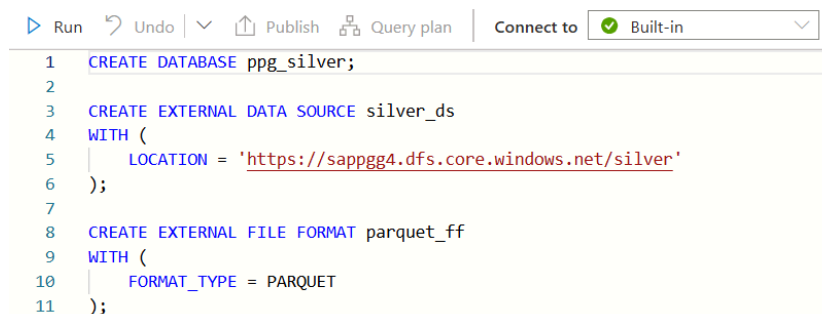
This picture shows the audit query for the sales_orders.csv file. The query counts the total rows, the distinct sales order IDs, the number of Open and Delivered orders, and the distinct customers and finished products. The result confirms 48 sales orders with 10 Open and 38 Delivered orders. This data is important for Q7, which traces how stockout risks affect open customer sales orders.

5.2.2 Silver Layer - Data Cleaning

The Silver Layer was implemented to improve the quality and consistency of the raw datasets stored in the Bronze Layer. Its primary purpose is to clean, standardize, and transform the data before it is used for further analysis in the Gold Layer.

Silver 0 - Setup

Initially, a setup script named Silver 0 – Setup was created to establish the Silver Layer environment. This script was responsible for creating the ppg_silver database and configuring the necessary external data source and file format required for storing transformed datasets in Parquet format.



```

1 CREATE DATABASE ppg_silver;
2
3 CREATE EXTERNAL DATA SOURCE silver_ds
4 WITH (
5     LOCATION = 'https://sappgg4.dfs.core.windows.net/silver'
6 );
7
8 CREATE EXTERNAL FILE FORMAT parquet_ff
9 WITH (
10     FORMAT_TYPE = PARQUET
11 );

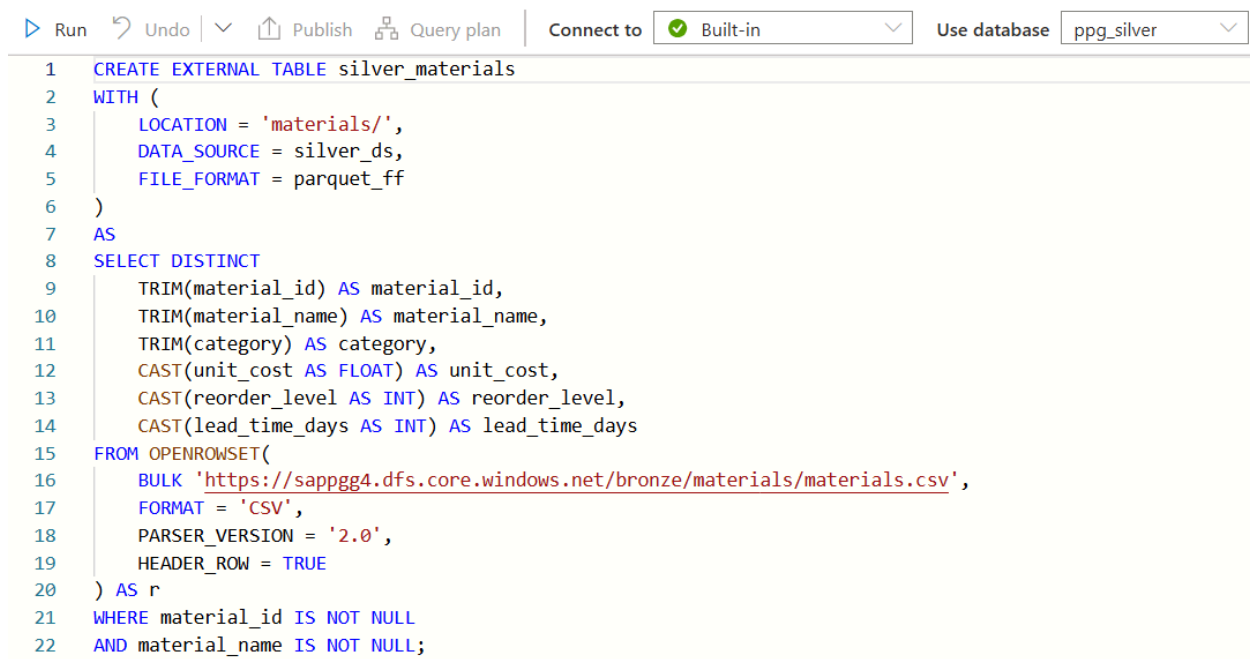
```

Figure 5.2.2.1: Silver Layer Setup Script Execution

Once the setup was completed, six CETAS (Create External Table As Select) scripts were developed to process the six datasets obtained from the Bronze Layer. Each script reads data from the Bronze container, applies the required transformations, and stores the cleaned output in the Silver container.

Silver Materials CETAS

The first transformation script was created for the Materials dataset. During this process, unnecessary spaces in text fields such as material identifiers, material names, and categories were removed using the TRIM() function. Numeric attributes including unit cost, reorder level, and lead time were converted into appropriate data types using the CAST() function. Duplicate records were also removed using the DISTINCT keyword, while records with missing material identifiers or material names were excluded from the final output.

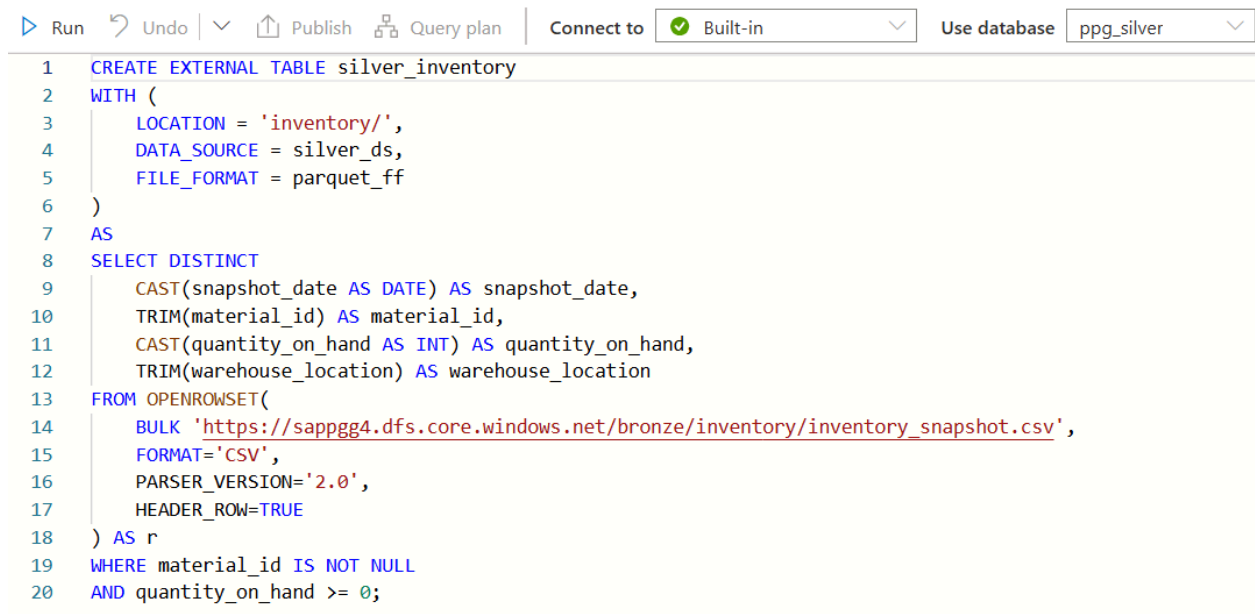


```
1 CREATE EXTERNAL TABLE silver_materials
2 WITH (
3     LOCATION = 'materials/',
4     DATA_SOURCE = silver_ds,
5     FILE_FORMAT = parquet_ff
6 )
7 AS
8 SELECT DISTINCT
9     TRIM(material_id) AS material_id,
10    TRIM(material_name) AS material_name,
11    TRIM(category) AS category,
12    CAST(unit_cost AS FLOAT) AS unit_cost,
13    CAST(reorder_level AS INT) AS reorder_level,
14    CAST(lead_time_days AS INT) AS lead_time_days
15 FROM OPENROWSET(
16     BULK 'https://sappgg4.dfs.core.windows.net/bronze/materials/materials.csv',
17     FORMAT = 'CSV',
18     PARSER_VERSION = '2.0',
19     HEADER_ROW = TRUE
20 ) AS r
21 WHERE material_id IS NOT NULL
22 AND material_name IS NOT NULL;
```

Figure 5.2.2.2: Materials Transformation Script

Silver Inventory CETAS

The second script processed the Inventory dataset. The snapshot date was converted into the DATE format, while inventory quantities were converted into integer values. In addition, records containing missing material identifiers or negative inventory quantities were filtered out to ensure that only valid inventory records were retained.

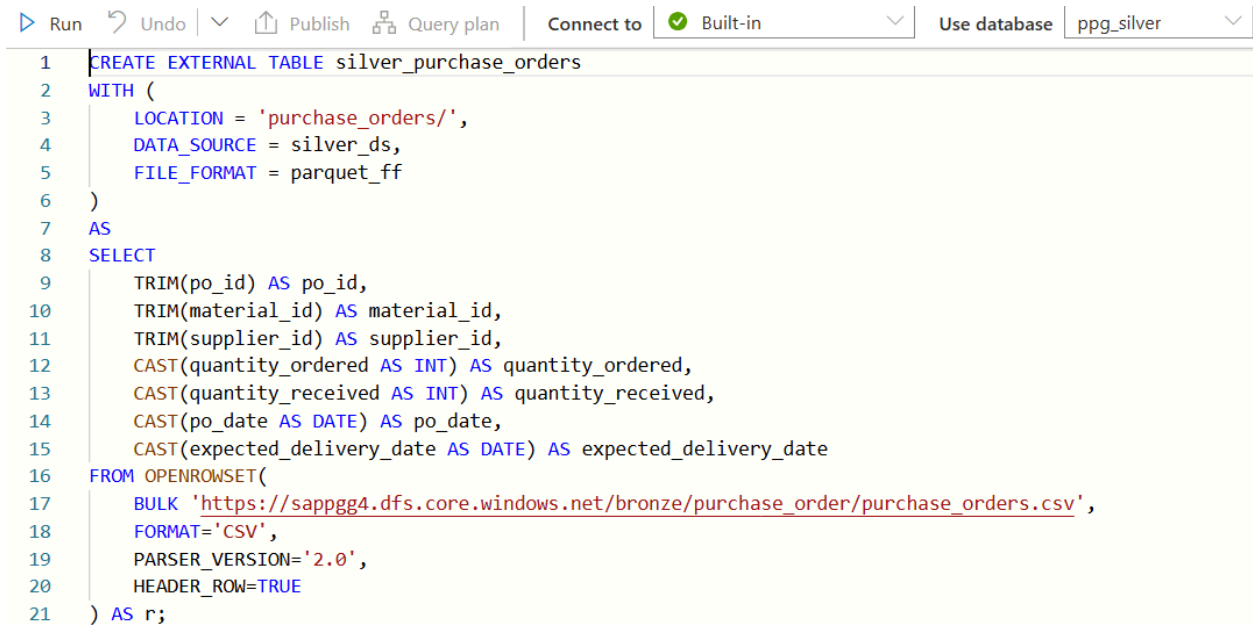


```
1 CREATE EXTERNAL TABLE silver_inventory
2 WITH (
3     LOCATION = 'inventory/',
4     DATA_SOURCE = silver_ds,
5     FILE_FORMAT = parquet_ff
6 )
7 AS
8 SELECT DISTINCT
9     CAST(snapshot_date AS DATE) AS snapshot_date,
10    TRIM(material_id) AS material_id,
11    CAST(quantity_on_hand AS INT) AS quantity_on_hand,
12    TRIM(warehouse_location) AS warehouse_location
13 FROM OPENROWSET(
14     BULK 'https://sappgg4.dfs.core.windows.net/bronze/inventory/inventory_snapshot.csv',
15     FORMAT='CSV',
16     PARSER_VERSION='2.0',
17     HEADER_ROW=TRUE
18 ) AS r
19 WHERE material_id IS NOT NULL
20 AND quantity_on_hand >= 0;
```

Figure 5.2.2.3: Inventory Transformation Script

Silver Purchase Orders CETAS

The Purchase Orders dataset was then transformed by standardizing text fields and converting quantity and date attributes into their appropriate data types. This ensures that purchase order information can be analyzed more effectively in later stages.



```

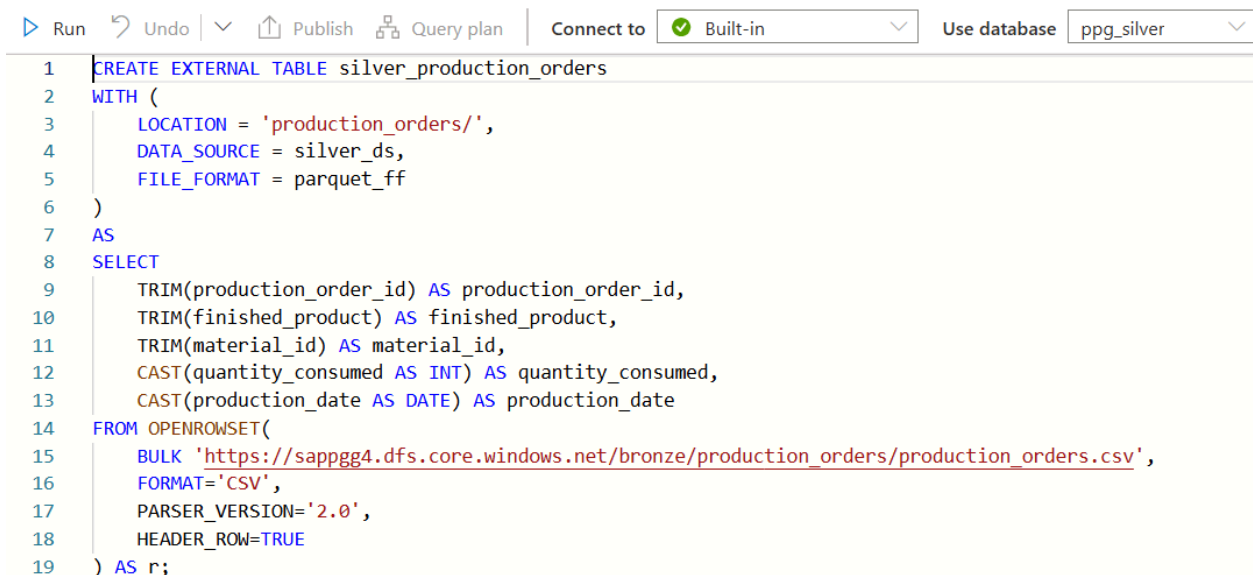
1 CREATE EXTERNAL TABLE silver_purchase_orders
2 WITH (
3     LOCATION = 'purchase_orders/',
4     DATA_SOURCE = silver_ds,
5     FILE_FORMAT = parquet_ff
6 )
7 AS
8 SELECT
9     TRIM(po_id) AS po_id,
10    TRIM(material_id) AS material_id,
11    TRIM(supplier_id) AS supplier_id,
12    CAST(quantity_ordered AS INT) AS quantity_ordered,
13    CAST(quantity_received AS INT) AS quantity_received,
14    CAST(po_date AS DATE) AS po_date,
15    CAST(expected_delivery_date AS DATE) AS expected_delivery_date
16 FROM OPENROWSET(
17     BULK 'https://sappgg4.dfs.core.windows.net/bronze/purchase_order/purchase_orders.csv',
18     FORMAT='CSV',
19     PARSER_VERSION='2.0',
20     HEADER_ROW=TRUE
21 ) AS r;

```

Figure 5.2.2.4: Purchase Orders Transformation Script

Silver Production Orders CETAS

For the Production Orders dataset, unnecessary spaces were removed from text fields and the quantity consumed and production date attributes were converted into suitable formats. This helps improve data consistency and supports future analytical processing.



```

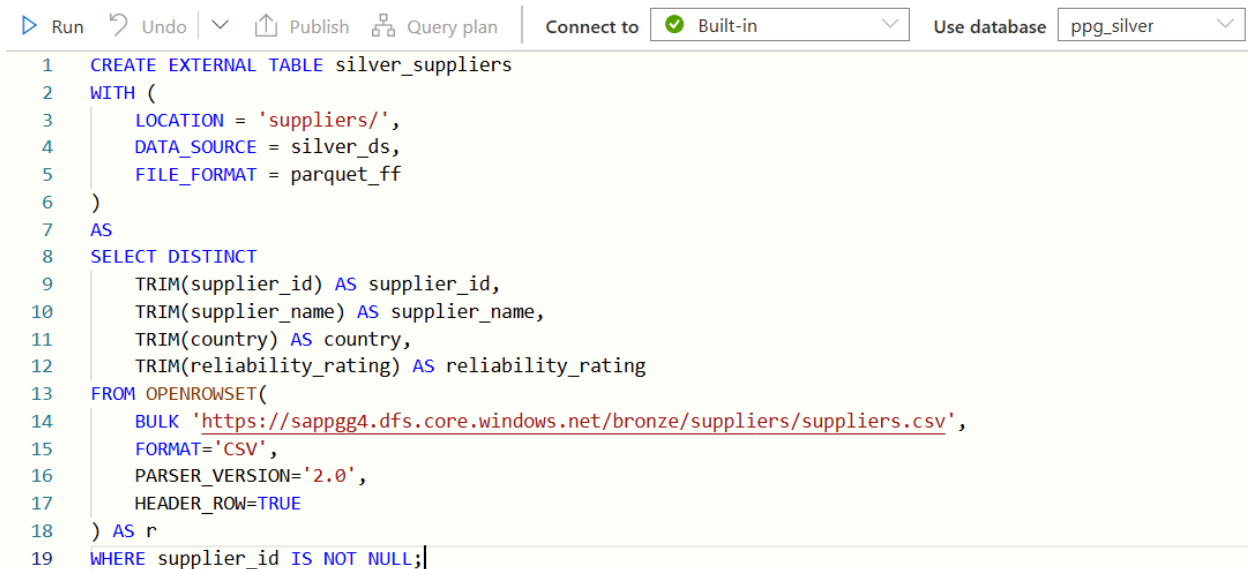
1 CREATE EXTERNAL TABLE silver_production_orders
2 WITH (
3     LOCATION = 'production_orders/',
4     DATA_SOURCE = silver_ds,
5     FILE_FORMAT = parquet_ff
6 )
7 AS
8 SELECT
9     TRIM(production_order_id) AS production_order_id,
10    TRIM(finished_product) AS finished_product,
11    TRIM(material_id) AS material_id,
12    CAST(quantity_consumed AS INT) AS quantity_consumed,
13    CAST(production_date AS DATE) AS production_date
14 FROM OPENROWSET(
15     BULK 'https://sappgg4.dfs.core.windows.net/bronze/production_orders/production_orders.csv',
16     FORMAT='CSV',
17     PARSER_VERSION='2.0',
18     HEADER_ROW=TRUE
19 ) AS r;

```

Figure 5.2.2.5: Production Orders Transformation Script

Silver Suppliers CETAS

The Suppliers dataset was cleaned by removing duplicate records and standardizing supplier-related information. Records with missing supplier identifiers were excluded to maintain data integrity.

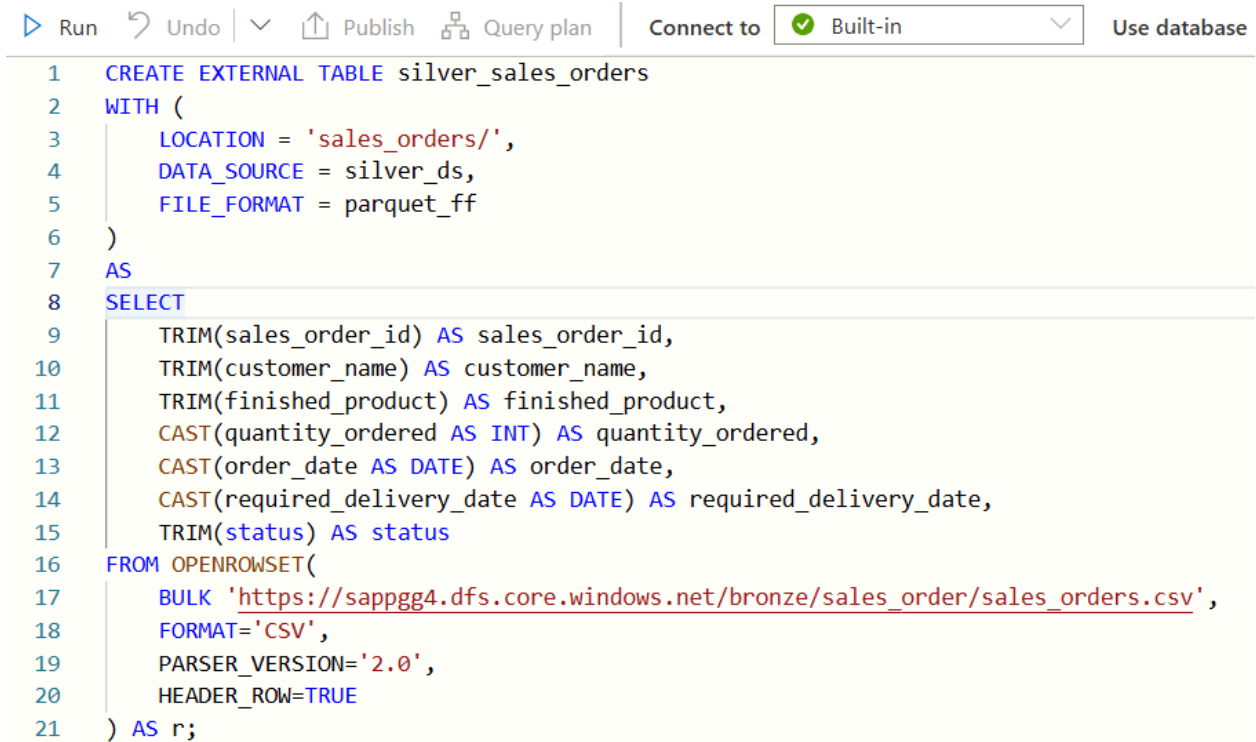


```
1 CREATE EXTERNAL TABLE silver_suppliers
2 WITH (
3     LOCATION = 'suppliers/',
4     DATA_SOURCE = silver_ds,
5     FILE_FORMAT = parquet_ff
6 )
7 AS
8 SELECT DISTINCT
9     TRIM(supplier_id) AS supplier_id,
10    TRIM(supplier_name) AS supplier_name,
11    TRIM(country) AS country,
12    TRIM(reliability_rating) AS reliability_rating
13 FROM OPENROWSET(
14     BULK 'https://sappgg4.dfs.core.windows.net/bronze/suppliers/suppliers.csv',
15     FORMAT='CSV',
16     PARSER_VERSION='2.0',
17     HEADER_ROW=TRUE
18 ) AS r
19 WHERE supplier_id IS NOT NULL;
```

Figure 5.2.2.6: Suppliers Dataset Transformation Script

Silver Sales Orders CETAS

Finally, the Sales Orders dataset was transformed by standardizing text fields and converting quantity and date attributes into the appropriate formats. This ensures that customer order information remains consistent and ready for analysis.



```
1 CREATE EXTERNAL TABLE silver_sales_orders
2 WITH (
3     LOCATION = 'sales_orders/',
4     DATA_SOURCE = silver_ds,
5     FILE_FORMAT = parquet_ff
6 )
7 AS
8 SELECT
9     TRIM(sales_order_id) AS sales_order_id,
10    TRIM(customer_name) AS customer_name,
11    TRIM(finished_product) AS finished_product,
12    CAST(quantity_ordered AS INT) AS quantity_ordered,
13    CAST(order_date AS DATE) AS order_date,
14    CAST(required_delivery_date AS DATE) AS required_delivery_date,
15    TRIM(status) AS status
16 FROM OPENROWSET(
17     BULK 'https://sappgg4.dfs.core.windows.net/bronze/sales_order/sales_orders.csv',
18     FORMAT='CSV',
19     PARSER_VERSION='2.0',
20     HEADER_ROW=TRUE
21 ) AS r;
```

Figure 5.2.2.7: Sales Orders Dataset Transformation Script

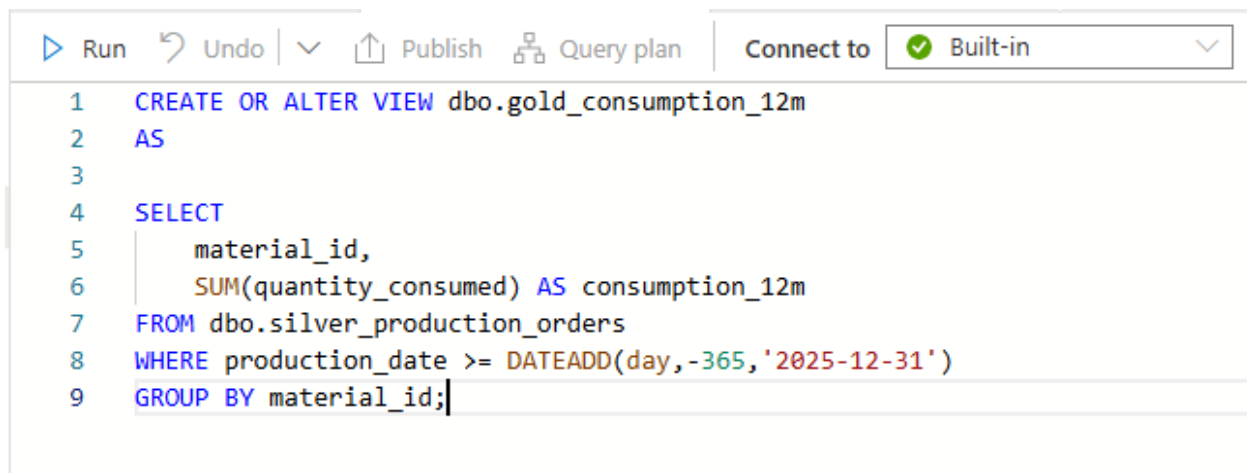
After all transformation activities were completed, the cleaned datasets were exported into the Silver container using the CETAS approach. The datasets were stored in Parquet format because it offers better storage efficiency and faster query performance compared to traditional CSV files. These cleaned datasets serve as the foundation for the Gold Layer, where business metrics, KPIs, and dashboard visualizations are generated.

5.2.3 Gold Layer - RA/MC Business Logic

This section describes the implementation of the Gold Layer business logic used to support inventory risk analysis and financial provisioning within the PPG Inventory Pipeline project. The Gold Layer integrates data from the Silver Layer and applies a series of business rules to identify materials that require inventory management attention. The implemented logic includes twelve-month consumption analysis, stock-on-hand aggregation, Recoverable Asset (RA) classification, Magna Carta (MC) Dead Stock identification, Magna Carta Expired material detection, and provision amount calculation. These business rules form the basis for answering business questions Q1 to Q5.

Twelve-Month Consumption Calculation

The SQL view `gold_consumption_12M` is displayed in figure below. Using manufacturing order information, the view determines the total amount of material consumed over the preceding 12 months. The analysis aggregates the amount utilized for each material using December 31, 2025, as the cutoff date. By comparing annual consumption to current stock levels, the resulting consumption value is then utilized to determine Recoverable Assets.

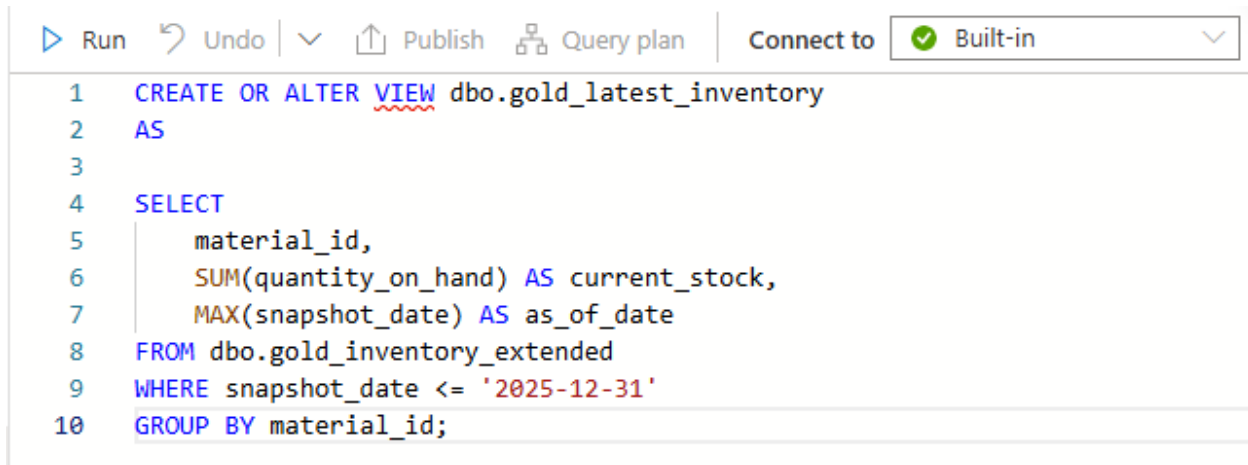


```
1 CREATE OR ALTER VIEW dbo.gold_consumption_12m
2 AS
3
4 SELECT
5     material_id,
6     SUM(quantity_consumed) AS consumption_12m
7 FROM dbo.silver_production_orders
8 WHERE production_date >= DATEADD(day, -365, '2025-12-31')
9 GROUP BY material_id;
```

Figure 5.2.3.1: Gold Consumption 12M Script

Latest Inventory Aggregation

This figure below shows the SQL view gold_latest_inventory. The view aggregates inventory records from the inventory snapshot dataset to determine the most recent stock-on-hand quantity for each commodity. Included are just inventory records from on or before the analysis cutoff date. For all inventory risk computations, the resulting stock quantity serves as the current inventory position.



```
1 CREATE OR ALTER VIEW dbo.gold_latest_inventory
2 AS
3
4 SELECT
5     material_id,
6     SUM(quantity_on_hand) AS current_stock,
7     MAX(snapshot_date) AS as_of_date
8 FROM dbo.gold_inventory_extended
9 WHERE snapshot_date <= '2025-12-31'
10 GROUP BY material_id;
```

Figure 5.2.3.2: Gold Latest Consumption Script

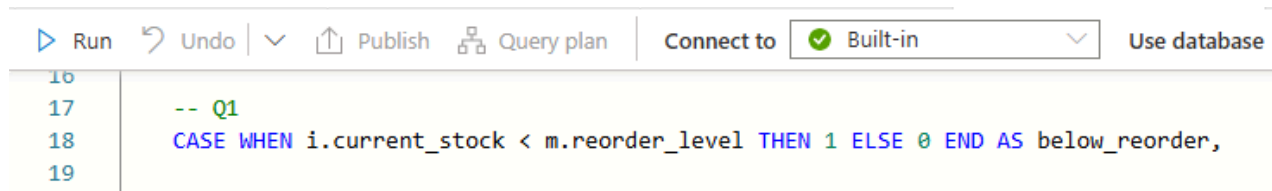
Inventory Risk Classification Flags

The SQL view fact_inventory_status, which includes the primary business rules for inventory risk classification, is displayed in this figure. The view creates the inventory status indicators needed for business questions Q1 through Q5 by combining material master data, inventory balances, consumption history, and expiry information.

Q1 - Below Reorder Flag

This flag identifies materials whose current inventory level is lower than the predefined reorder level specified in the material master data. Materials meeting this condition may require replenishment to avoid future shortages.

The following logic was implemented:



```
16
17      -- Q1
18      CASE WHEN i.current_stock < m.reorder_level THEN 1 ELSE 0 END AS below_reorder,,
19
```

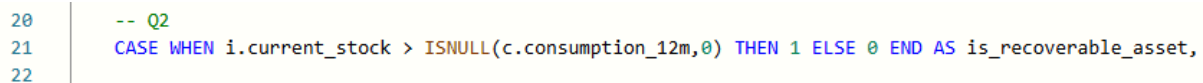
Figure 5.2.3.3: Gold Q1 Fact Inventory Status Script

A value of 1 indicates that the material is below the reorder level, while 0 indicates that the current stock level remains above the minimum threshold.

Q2 - Recoverable Asset Flag

This flag identifies materials whose current stock quantity exceeds total consumption during the previous twelve months. Such materials are considered Recoverable Assets because the inventory available is greater than historical demand.

The following logic was implemented:



```
20
21      -- Q2
22      CASE WHEN i.current_stock > ISNULL(c.consumption_12m,0) THEN 1 ELSE 0 END AS is_recoverable_asset,
```

Figure 5.2.3.4: Gold Q2 Fact Inventory Status Script

A value of 1 indicates that the material has excess inventory and qualifies as a Recoverable Asset, while 0 indicates that the stock level is within the expected consumption range.

Q3 - Magna Carta(MC) Dead Stock Flag

This flag identifies materials that have not been consumed for at least nine months and do not have a recorded expiry date. These materials are considered inactive inventory and may require further review for disposal or write-off.

The following logic was implemented:

```

22
23      -- Q3
24      CASE
25      |   WHEN (l.last_consumed_date IS NULL OR DATEDIFF(MONTH, l.last_consumed_date, '2025-12-31') >= 9)
26      |   |   AND e.lot_expiry_date IS NULL
27      |   |   THEN 1 ELSE 0
28      |   END AS is_mc_dead_stock,
29

```

Figure 5.2.3.5: Gold Q3 Fact Inventory Status Script

A value of 1 indicates that the material satisfies both Dead Stock conditions:

1. No consumption activity for at least nine months.
2. No lot expiry date available.

A value of 0 indicates that one or both conditions were not met.

Q4 - Magna Carta(MC) Expired Flag

This flag identifies materials whose lot expiry date has passed before the analysis cutoff date. Expired materials are classified separately because they can no longer be used for production or operational purposes.

The following logic was implemented:

```

29
30      -- Q4
31      CASE
32      |   WHEN e.lot_expiry_date IS NOT NULL AND e.lot_expiry_date < '2025-12-31'
33      |   |   THEN 1 ELSE 0
34      |   END AS is_mc_expired,

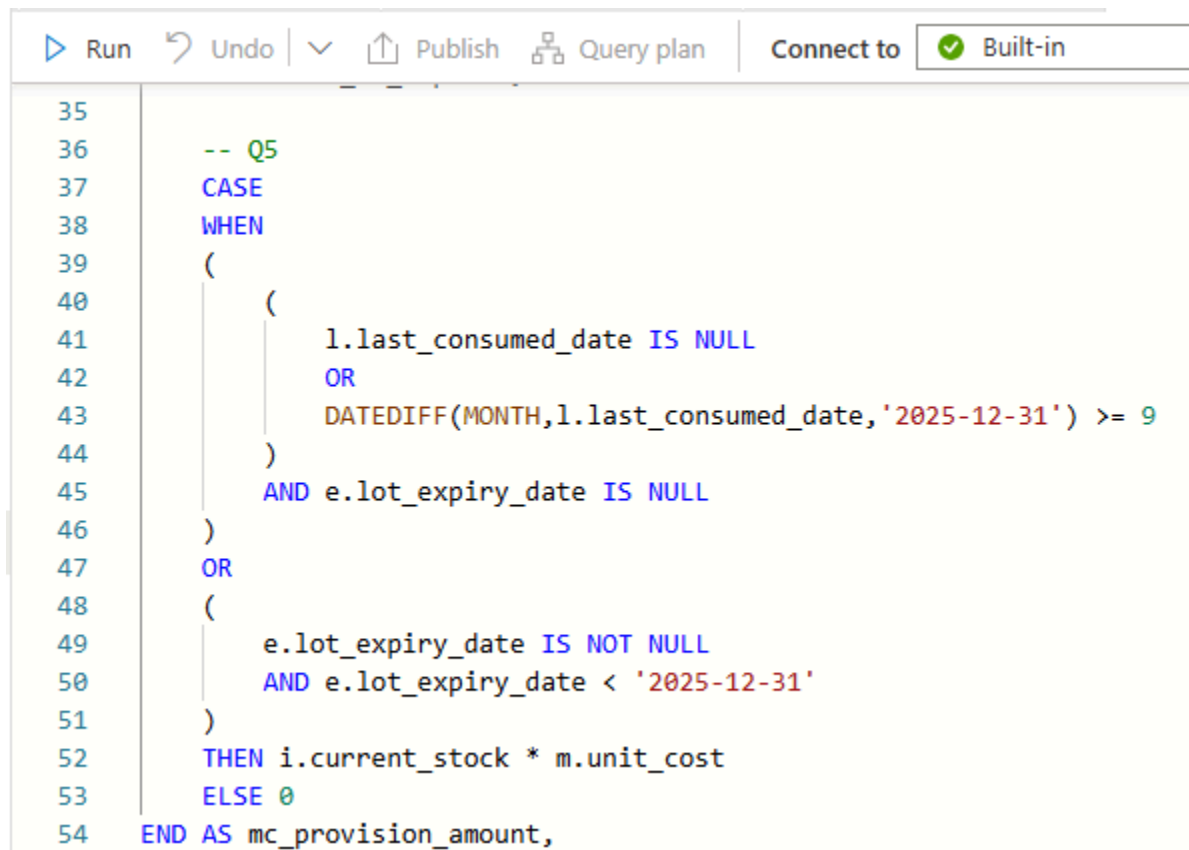
```

Figure 5.2.3.6: Gold Q4 Fact Inventory Status Script

A value of 1 indicates that the material expiry date occurred before the analysis cutoff date, while 0 indicates that the material has not expired.

Magna Carta(MC) Provision Calculation

This picture shows the financial provisioning logic used for MC Dead Stock and MC Expired materials. The provision amount is calculated by multiplying the current stock quantity by the material unit cost. This calculation estimates the financial write-down required for obsolete inventory.

The image shows a screenshot of a SQL query editor interface. At the top, there is a toolbar with icons for 'Run', 'Undo', 'Publish', 'Query plan', and 'Connect to'. The 'Connect to' dropdown is set to 'Built-in'. The main area displays a SQL script with line numbers 35 through 54. The script is a CASE WHEN statement that calculates a provision amount based on inventory status and lot expiry date. The logic is as follows: if the last consumed date is NULL or the date difference from 2025-12-31 is greater than or equal to 9 months, and the lot expiry date is NULL, then the provision amount is the current stock multiplied by the unit cost. Otherwise, the provision amount is 0.

```
35
36  -- Q5
37  CASE
38  WHEN
39  (
40  (
41      l.last_consumed_date IS NULL
42      OR
43      DATEDIFF(MONTH,l.last_consumed_date,'2025-12-31') >= 9
44  )
45      AND e.lot_expiry_date IS NULL
46  )
47  OR
48  (
49      e.lot_expiry_date IS NOT NULL
50      AND e.lot_expiry_date < '2025-12-31'
51  )
52  THEN i.current_stock * m.unit_cost
53  ELSE 0
54  END AS mc_provision_amount,
```

Figure 5.2.3.7: Gold Q5 Fact Inventory Status Script

5.2.4 Gold Layer - D1 Script, Q6/Q7 Sales Impact

D1 Data Extension – Adding lot_expiry_date

The original inventory dataset did not contain the lot_expiry_date attribute required for Magna Carta Expired (Q4) analysis. To solve this issue, a Gold Layer view named

gold_inventory_extended was created to extend the inventory data with expiry date information. The generated expiry dates were assigned according to the following business rules:

- MAT001 and MAT005 were assigned an expiry date of 2025-06-01 to represent expired inventory.
- MAT016 to MAT020 were assigned NULL values to represent materials that do not require expiry tracking.
- All remaining materials were assigned an expiry date of 2026-12-31 to represent valid inventory.

```
CREATE OR ALTER VIEW dbo.gold_inventory_extended
AS
SELECT
    snapshot_date,
    material_id,
    quantity_on_hand,
    warehouse_location,
    CASE
        WHEN material_id IN ('MAT001','MAT005')
        THEN CAST('2025-06-01' AS DATE)

        WHEN material_id IN
        (
            'MAT016',
            'MAT017',
            'MAT018',
            'MAT019',
            'MAT020'
        )
        THEN NULL

        ELSE CAST('2026-12-31' AS DATE)
    END AS lot_expiry_date
FROM ppg_silver.dbo.silver_inventory;
```

Probability Distribution Explanation

Packaging materials were assigned NULL expiry dates because expiry tracking is not applicable. Two materials were intentionally assigned expired dates to support Q4 testing, while remaining materials were assigned future expiry dates representing valid inventory.

Expiry Category	Materials	Count	Percentage
Expired	MAT001, MAT005	2	10%
No Expiry Tracking	MAT016-MAT020	5	25%
Future Expiry	Remaining Materials	13	65%

Q6-unfulfilled production query logic

The objective of Q6 was to determine whether any production orders could not be fulfilled due to insufficient inventory. Production orders were joined with the Gold Layer inventory status table. The quantity required by each production order was compared against required by each production order was compared against the available stock quantity. Production orders was classified as unfulfilled when $\text{Required Quantity} > \text{Current Stock}$.

```

CREATE OR ALTER VIEW dbo.gold_unfulfilled_production
AS

SELECT

    p.production_order_id,
    p.finished_product,
    p.material_id,

    p.quantity_consumed,

    f.current_stock,

    (p.quantity_consumed - f.current_stock)
AS shortfall

FROM ppg_silver.dbo.silver_production_orders p

INNER JOIN dbo.fact_inventory_status f
ON p.material_id = f.material_id

WHERE p.quantity_consumed > f.current_stock;

```

Q7 At-risk sales orders tracing query

The objective of Q7 was to identify which open customer sales orders would be impacted by materials classified as stockout risk. The analysis traced the relationship between customer orders, production orders, and inventory materials by linking sales orders to production orders through the finished product and subsequently linking production orders to the inventory status table. Materials flagged as stockout risk were then used to determine which customer orders could potentially be affected.

The relationship flow :

Sales Orders → Finished Product → Production Orders → Material → Stockout Risk

This analysis successfully identified the affected sales orders and customers associated with inventory stockout risks, enabling proactive inventory and customer order management.


```

CREATE OR ALTER VIEW dbo.gold_unfulfilled_production
AS

SELECT

p.production_order_id,
p.finished_product,
p.material_id,

p.quantity_consumed,

f.current_stock,

(p.quantity_consumed - f.current_stock)
AS shortfall

FROM ppg_silver.dbo.silver_production_orders p

INNER JOIN dbo.fact_inventory_status f
ON p.material_id = f.material_id

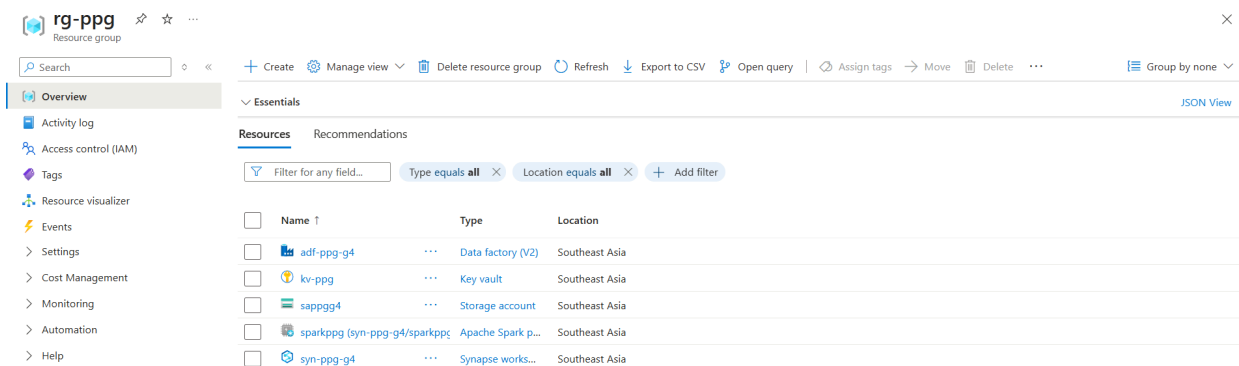
WHERE p.quantity_consumed > f.current_stock;

```

5.3 Interfaces of System Main Functions

5.3.1 Bronze Layer - Azure Setup & Ingestion

1. Azure Portal : Resource Group rg-ppg showing all 6 resources

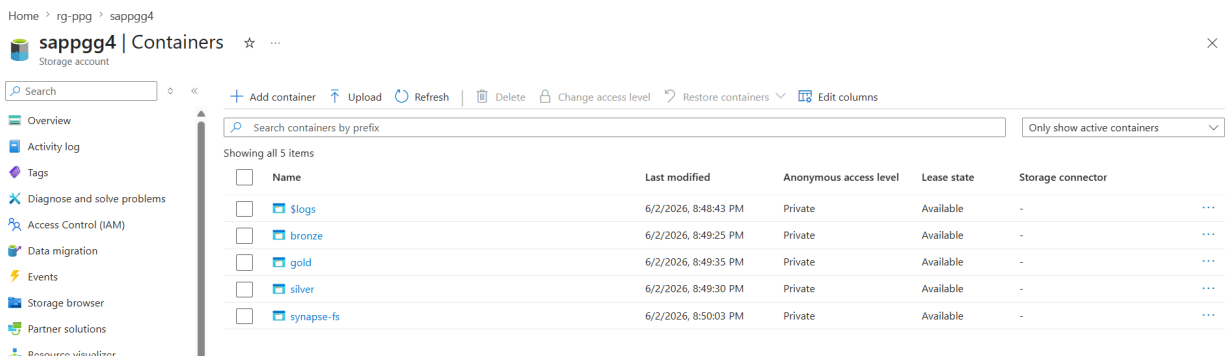


The screenshot shows the Azure Portal interface for the Resource Group 'rg-ppg'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Cost Management, Monitoring, Automation, and Help. The main area displays the 'Essentials' section with a 'Resources' tab selected. A table lists the resources within the group, all located in Southeast Asia.

Name	Type	Location
adf-ppg-g4	Data factory (V2)	Southeast Asia
kv-ppg	Key vault	Southeast Asia
sappg4	Storage account	Southeast Asia
sparkppg (syn-ppg-g4/sparkppg)	Apache Spark p...	Southeast Asia
syn-ppg-g4	Synapse works...	Southeast Asia

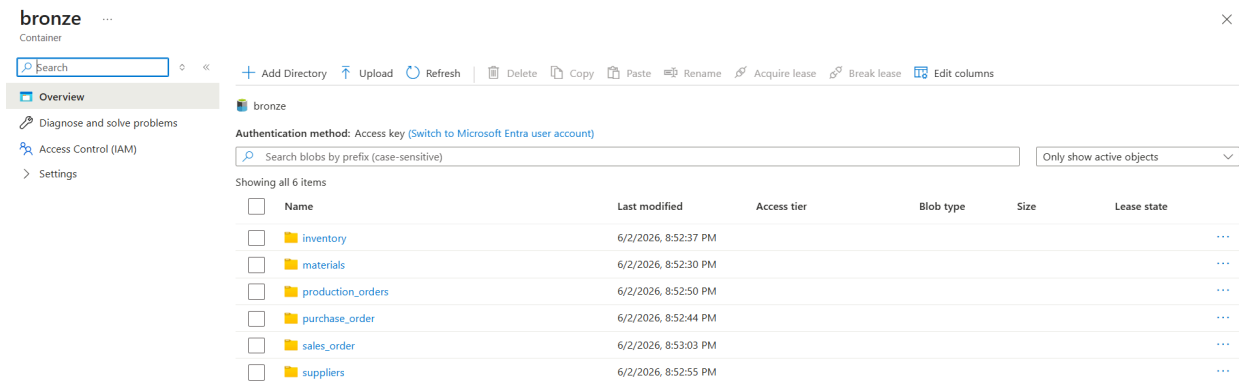
This picture shows the Azure Portal view of the resource group rg-ppg, which contains all six Azure resources used in this project. The resources include the Data Factory (adf-ppg-g4), Key Vault (kv-ppg), Storage Account (sappgg4), Apache Spark pool (sparkppg), and Synapse workspace (syn-ppg-g4). All resources are deployed in the Southeast Asia region. Grouping them under one resource group makes it easier to manage, monitor, and control access to the whole system

2. Storage Account : 4 containers listed (bronze, silver, gold, synapse-fs)



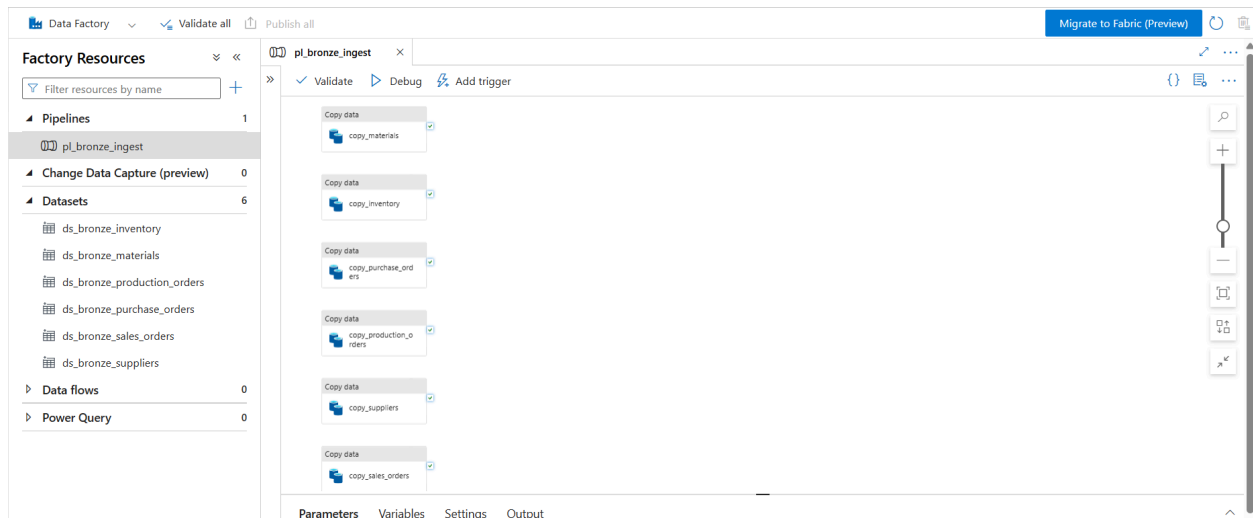
This picture shows the storage account sappgg4 with all five containers listed: \$logs, bronze, gold, silver, and synapse-fs. Each container is set to private access for security. The bronze, silver, and gold containers represent the three Medallion Architecture layers, while synapse-fs is used by the Synapse workspace. This structure ensures that data at each stage of the pipeline is stored separately and is easy to organise.

3. Bronze container : 6 subfolders with CSV files uploaded



This picture shows the inside of the bronze container, which has six subfolders, one for each source dataset: inventory, materials, production_orders, purchase_order, sales_order, and suppliers. Each subfolder stores its own raw CSV file. This folder structure keeps the raw data clearly separated by dataset and makes it easier to reference each file in later pipeline steps.

4. ADF Studio : pl_bronze_ingest pipeline canvas with 6 Copy activities



This picture shows the Azure Data Factory Studio with the pl_bronze_ingest pipeline open on the canvas. The pipeline contains six Copy Data activities (copy_materials, copy_inventory, copy_purchase_orders, copy_production_orders, copy_suppliers, and copy_sales_orders) and uses six datasets (ds_bronze_inventory, ds_bronze_materials, and so on). This single pipeline is the main tool used to load all six raw CSV files into the Bronze container in one run.

5. ADF Monitor : Pipeline run showing all 6 activities green (success)

Data Factory Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 1
 - pl_bronze_ingest
- Change Data Capture (preview) 0
- Datasets 6
 - ds_bronze_inventory
 - ds_bronze_materials
 - ds_bronze_production_orders
 - ds_bronze_purchase_orders
 - ds_bronze_sales_orders
 - ds_bronze_suppliers
- Data flows 0
- Power Query 0

pl_bronze_ingest

Validate Debug Add trigger

- Copy data ✓
copy_materials
- Copy data ✓
copy_inventory
- Copy data ✓
copy_purchase_orders
- Copy data ✓
copy_production_orders
- Copy data ✓
copy_suppliers
- Copy data ✓
copy_sales_orders

Activity name ↑↓	Activity st... ↑↓	Activit... ↑↓	Run start ↑↓	Duration ↑↓	Integration runtime ↑↓	User prop... ↑↓
copy_suppliers	✓ Succeeded	Copy data	6/8/2026, 2:26:28 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_sales_orders	✓ Succeeded	Copy data	6/8/2026, 2:26:28 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_inventory	✓ Succeeded	Copy data	6/8/2026, 2:26:28 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_purchase_orders	✓ Succeeded	Copy data	6/8/2026, 2:26:28 PM	19s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_materials	✓ Succeeded	Copy data	6/8/2026, 2:26:28 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_production_orders	✓ Succeeded	Copy data	6/8/2026, 2:26:28 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)	

This picture shows the ADF Monitor page after running the pl_bronze_ingest pipeline. All six Copy Data activities show a green Succeeded status, with run durations between 18 and 20 seconds each. This confirms that every CSV file was copied into the Bronze container without any error, proving that the ingestion process is working correctly and is ready for the next stage of the pipeline.

6. Synapse Studio : All 6 audit query results (one screenshot per query)

Audit Query 1 - Materials

Audit Query 1 - mat...

Run Undo Publish Query plan Connect to Built-in Use database master

```
1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT material_id) AS distinct_material_ids,
4     SUM(CASE WHEN material_id IS NULL THEN 1 ELSE 0 END) AS null_material_ids,
5     SUM(CASE WHEN material_name IS NULL THEN 1 ELSE 0 END) AS null_material_names,
6     SUM(CASE WHEN unit_cost IS NULL THEN 1 ELSE 0 END) AS null_unit_costs,
7     SUM(CASE WHEN reorder_level IS NULL THEN 1 ELSE 0 END) AS null_reorder_levels,
8     COUNT(DISTINCT category) AS distinct_categories
9 FROM OPENROWSET(
10     BULK 'https://sappgg4.dfs.core.windows.net/bronze/materials/materials.csv',
11     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
12 ) AS r;
```

Results Messages

View Table Chart Export results

Search

total_rows	distinct_material_ids	null_material_ids	null_material_names	null_unit_costs	null_reorder_levels	distinct_categories
20	20	0	0	0	0	4

00:00:08 Query executed successfully.

Audit Query 2 - inventory_snapshot

Audit Query 2 - inve...

Run Undo Publish Query plan Connect to Built-in Use database master

```
1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT material_id) AS distinct_material_ids,
4     COUNT(DISTINCT snapshot_date) AS distinct_snapshot_months,
5     SUM(CASE WHEN quantity_on_hand IS NULL THEN 1 ELSE 0 END) AS null_quantity,
6     SUM(CASE WHEN quantity_on_hand < 0 THEN 1 ELSE 0 END) AS negative_quantity_rows,
7     COUNT(DISTINCT warehouse_location) AS distinct_warehouses
8 FROM OPENROWSET(
9     BULK 'https://sappgg4.dfs.core.windows.net/bronze/inventory/inventory_snapshot.csv',
10     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
11 ) AS r;
```

Results Messages

View Table Chart Export results

Search

total_rows	distinct_material_ids	distinct_snapshot_months	null_quantity	negative_quantity_rows	distinct_warehouses
240	20	12	0	0	5

00:00:08 Query executed successfully.

Audit Query 3 - purchase_orders

Audit Query 3 - pur...

Run Undo Publish Query plan Connect to Built-in Use database master

```
1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT po_id) AS distinct_po_ids,
4     SUM(CASE WHEN actual_delivery_date = '' THEN 1 ELSE 0 END) AS open_undelivered_pos,
5     SUM(CASE WHEN CAST(quantity_received AS INT) > CAST(quantity_ordered AS INT)
6         THEN 1 ELSE 0 END) AS over_delivered_rows,
7     SUM(CASE WHEN CAST(quantity_received AS INT) < CAST(quantity_ordered AS INT)
8         AND actual_delivery_date != ''
9         THEN 1 ELSE 0 END) AS partial_delivery_rows
10 FROM OPENROWSET(
11     BULK 'https://sappgg4.dfs.core.windows.net/bronze/purchase_order/purchase_orders.csv',
12     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
13 ) AS r;
```

Results Messages

View Table Chart Export results

Search

total_rows	distinct_po_ids	open_undelivered_pos	over_delivered_rows	partial_delivery_rows
40	40	0	0	7

00:00:00 Query executed successfully.

Audit Query 4 - production_orders

Audit Query 4 - pro...

Run Undo Publish Query plan Connect to Built-in Use database master

```
1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT production_order_id) AS distinct_production_orders,
4     COUNT(DISTINCT material_id) AS distinct_materials_consumed,
5     COUNT(DISTINCT finished_product) AS distinct_finished_products,
6     SUM(CASE WHEN quantity_consumed IS NULL THEN 1 ELSE 0 END) AS null_quantity_consumed,
7     MIN(production_date) AS earliest_production_date,
8     MAX(production_date) AS latest_production_date
9 FROM OPENROWSET(
10     BULK 'https://sappgg4.dfs.core.windows.net/bronze/production_orders/production_orders.csv',
11     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
12 ) AS r;
```

Results Messages

View Table Chart Export results

Search

total_rows	distinct_production_or...	distinct_materials_cons...	distinct_finished_prod...	null_quantity_consu...	earliest_production_date	latest_production_date
336	84	15	10	0	2025-01-15	2025-12-05

00:00:00 Query executed successfully.

Audit Query 5 - suppliers

Audit Query 5 - sup...

Run Undo Publish Query plan Connect to Built-in Use database master

```

1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT supplier_id) AS distinct_supplier_ids,
4     SUM(CASE WHEN supplier_id IS NULL THEN 1 ELSE 0 END) AS null_supplier_ids,
5     COUNT(DISTINCT country) AS distinct_countries,
6     COUNT(DISTINCT reliability_rating) AS distinct_ratings
7 FROM OPENROWSET(
8     BULK 'https://sappgg4.dfs.core.windows.net/bronze/suppliers/suppliers.csv',
9     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
10 ) AS r;

```

Results Messages

View Table Chart Export results

Search

total_rows	distinct_supplier_ids	null_supplier_ids	distinct_countries	distinct_ratings
10	10	0	7	3

00:00:00 Query executed successfully.

Audit Query 6 - sales_orders

Audit Query 6 - sale...

Run Undo Publish Query plan Connect to Built-in Use database master

```

1 SELECT
2     COUNT(*) AS total_rows,
3     COUNT(DISTINCT sales_order_id) AS distinct_order_ids,
4     SUM(CASE WHEN status = 'Open' THEN 1 ELSE 0 END) AS open_orders,
5     SUM(CASE WHEN status = 'Delivered' THEN 1 ELSE 0 END) AS delivered_orders,
6     COUNT(DISTINCT customer_name) AS distinct_customers,
7     COUNT(DISTINCT finished_product) AS distinct_products,
8     MIN(required_delivery_date) AS earliest_delivery,
9     MAX(required_delivery_date) AS latest_delivery
10 FROM OPENROWSET(
11     BULK 'https://sappgg4.dfs.core.windows.net/bronze/sales_order/sales_orders.csv',
12     FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
13 ) AS r;

```

Results Messages

View Table Chart Export results

Search

total_rows	distinct_order_ids	open_orders	delivered_orders	distinct_customers	distinct_products	earliest_delivery	latest_delivery
48	48	10	38	15	10	2025-02-04	2026-02-01

00:00:00 Query executed successfully.

This picture shows the six audit queries run in Synapse Studio to verify the data inside the Bronze container. Each query reads one CSV file using OPENROWSET and counts the rows, checks for null values, and inspects key columns. The results confirm: 20 materials with 4 categories, 240 inventory rows across 12 months, 40 purchase orders, 336 production orders, 10 suppliers from 7 countries, and 48 sales orders (10 Open and 38 Delivered). All results match the expected values, which proves that the Bronze Layer ingestion is complete and accurate.

5.3.2 Silver Layer Interfaces

The implementation of the Silver Layer was verified through Azure Synapse Studio and Azure Data Lake Storage Gen2. Azure Synapse Studio was used to execute the transformation scripts, while Azure Storage was used to confirm that the transformed datasets were successfully generated and stored.

After the CETAS scripts were executed successfully, six folders were created within the Silver container. Each folder represents a cleaned version of the corresponding Bronze Layer dataset, namely Materials, Inventory, Purchase Orders, Production Orders, Suppliers, and Sales Orders.

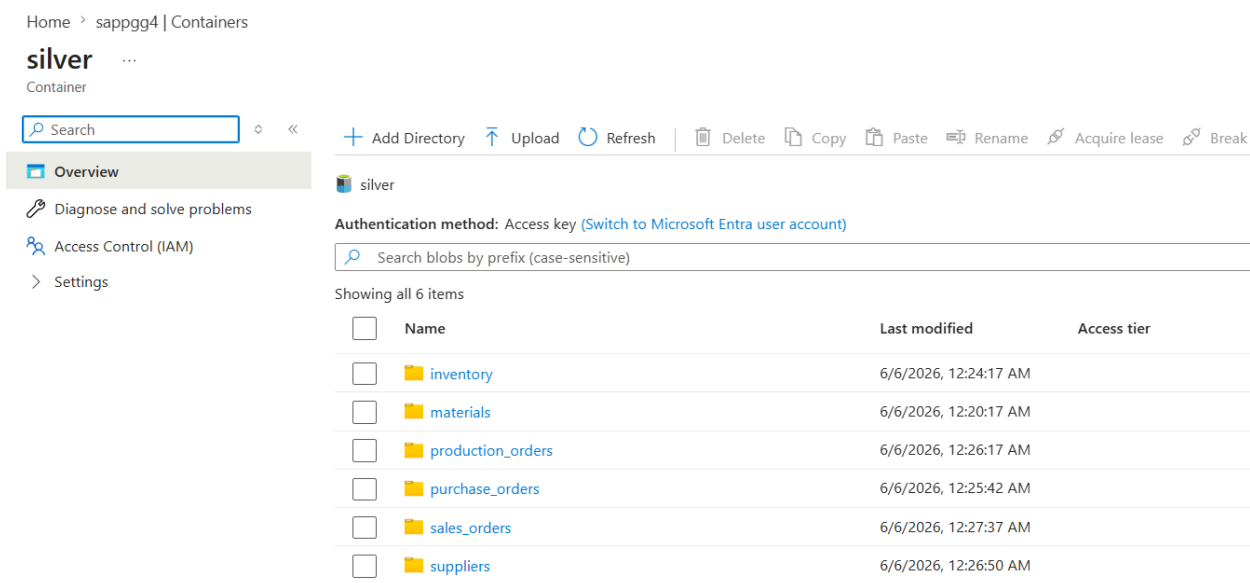


Figure 5.3.2.1: Silver Container Structure in Azure Data Lake Storage

Figure 5.3.2.1 shows the structure of the Silver container where the transformed datasets are stored. Organizing the datasets into separate folders improves data management and makes it easier for future processing activities.

Within each folder, the transformed datasets were generated as Parquet files. These files contain the cleaned and standardized data produced during the Silver Layer transformation process.

silver > materials

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive) Only show active objects

Showing all 2 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[-]					
☐	 E2C0DF48-6B30-40E3-B7C6-20BD11D247ED_31_0-1.parquet	6/6/2026, 12:20:22 AM	Hot (Inferred)	Block blob	3.1 KiB	Available
☐	 -	6/6/2026, 12:20:17 AM	Hot (Inferred)	Block blob	0	Available

Figure 5.3.2.2: Example of Generated Parquet File in the Silver Layer

The successful creation of the Parquet files indicates that the data transformation process was completed successfully. These datasets are then used as the primary input source for the Gold Layer analytical processing stage.

5.3.3 Gold RA/MC Interfaces

This section presents the outputs generated from the Gold Layer business logic implemented in Azure Synapse Analytics. The interfaces demonstrate the results of the inventory risk classification and financial provisioning analysis used to answer business questions Q1 to Q5. The query results were obtained from the fact_inventory_status view, which contains the final inventory status information generated after applying the Recoverable Assets (RA) and Magna Carta (MC) business rules. In addition, the storage structure of the Gold Layer dataset in Azure Data Lake Storage is also presented.

Q1 Query Result (materials below reorder level)

Run Undo Publish Query plan Connect to Built-in

```

1 SELECT COUNT(*) AS below_reorder_count
2 FROM dbo.fact_inventory_status
3 WHERE below_reorder=1;
4

```

Results Messages

View Table Chart Export results

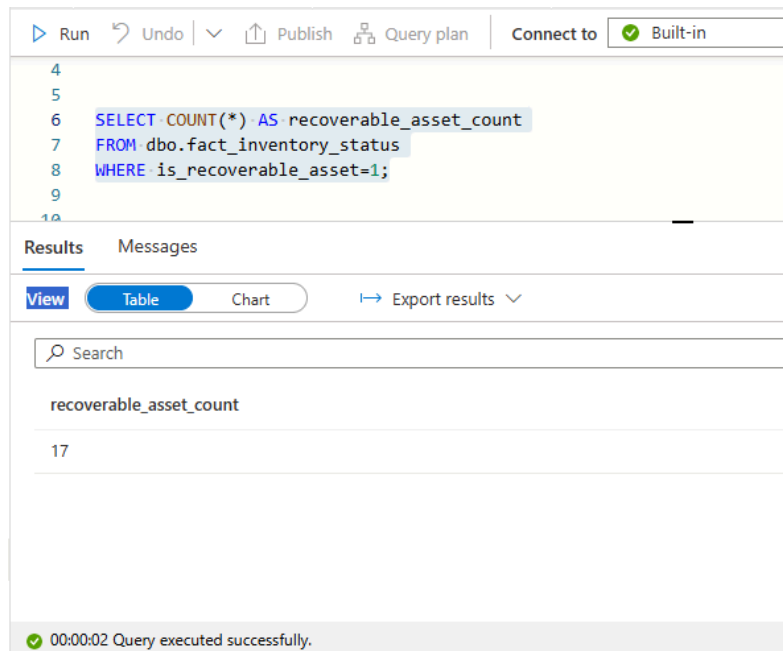
Search

below_reorder_count
0

Figure 5.3.3.1: Query Result for Materials Below Reorder Level

Records from the fact_inventory_status view when the below_reorder flag is equal to 1 are filtered by the query. This business rule compares each material's current stock level to the predetermined reorder level. As of the analysis cutoff date, no materials were below their reorder level, according to the result, which returned no data.

Q2 Query Result (Recoverable Assets list)



```
4
5
6 SELECT COUNT(*) AS recoverable_asset_count
7 FROM dbo.fact_inventory_status
8 WHERE is_recoverable_asset=1;
9
10
```

recoverable_asset_count
17

00:00:02 Query executed successfully.

Figure 5.3.3.2: Query Result for Recoverable Assets

Materials are considered Recoverable Assets when the current stock quantity exceeds total consumption during the previous twelve months. These materials represent excess inventory relative to historical demand. The result identified 17 materials that met the Recoverable Asset criteria.

Q3 Query Result (MC Dead Stock materials)

Run Undo | Publish Query plan | Connect to Built-in Use database ppg_gold

```
12 SELECT
13 material_id,
14 lot_expiry_date,
15 last_consumed_date,
16 is_mc_dead_stock
17 FROM dbo.fact_inventory_status
18 WHERE is_mc_dead_stock = 1;
```

Results Messages

View Table Chart Export results

Search

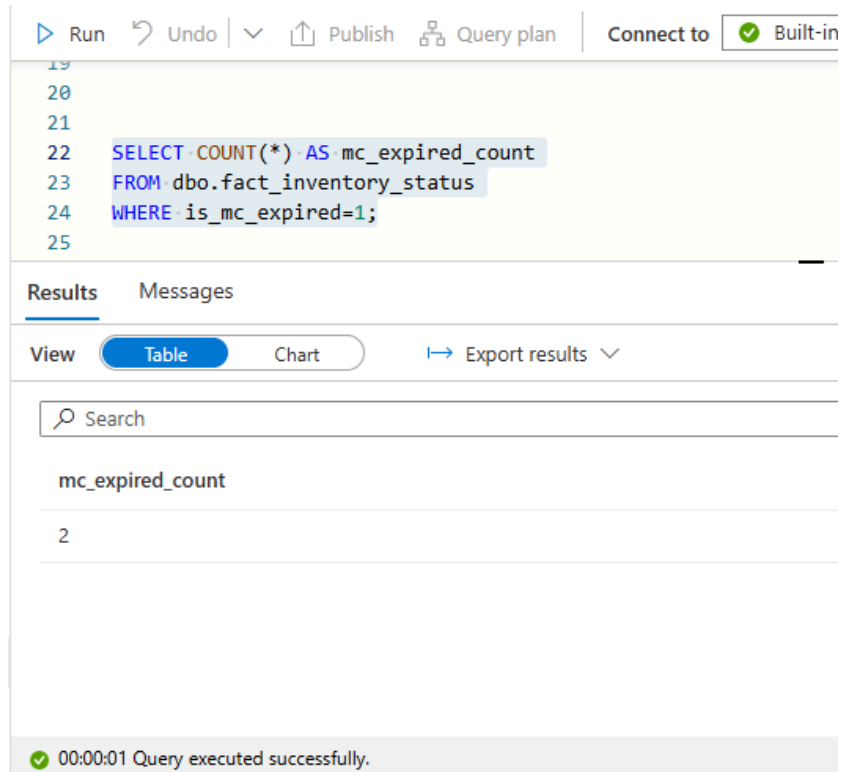
material_id	lot_expiry_date	last_consumed_date	is_mc_dead_stock
MAT019	(NULL)	(NULL)	1

00:00:00 Query executed successfully.

Figure 5.3.3.3: Query Result for MC Dead Stock Materials

Materials are considered Dead Stock when they have not been consumed for at least nine months and do not have a recorded lot expiry date. The result identified MAT019 as the only material meeting both conditions.

Q4 Query Result (MC Expired materials)



The screenshot displays a SQL query execution interface. The query is as follows:

```
19  
20  
21  
22 SELECT COUNT(*) AS mc_expired_count  
23 FROM dbo.fact_inventory_status  
24 WHERE is_mc_expired=1;  
25
```

The interface includes a toolbar with buttons for Run, Undo, Publish, Query plan, and Connect to. The Results tab is active, showing a table with one column, `mc_expired_count`, and one row with the value 2. A status bar at the bottom indicates the query executed successfully in 00:00:01.

mc_expired_count
2

Figure 5.3.3.4: Query Result for MC Expired Materials

Materials are considered expired when their lot expiry date occurs before the analysis cutoff date of 31 December 2025. The result identified two expired materials.

Q5 Query Result (total provision amount)

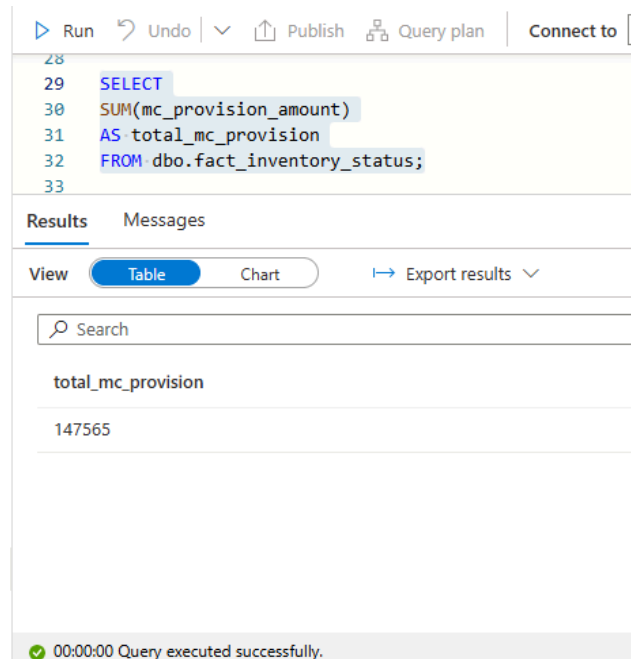


Figure 5.3.3.5: Query Result for Total MC Provision Amount

The provision amount represents the estimated financial write-down value for materials classified as Magna Carta Dead Stock or Magna Carta Expired. The result shows a total provision amount of RM147,565.

Azure Portal – Gold Container

gold > fact_inventory_status						
Authentication method: Access key (Switch to Microsoft Entra user account)						
Search blobs by prefix (case-sensitive)						Only show active objects
Showing all 1 items						
☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[..]					...
☐	fact_inventory_status.csv	6/8/2026, 4:12:51 PM	Hot (Inferred)	Block blob	2.3 KiB	Available

Figure 5.3.3.6: Gold Layer Parquet Files Stored in Azure Data Lake Storage

The fact_inventory_status dataset is stored in Parquet format within the Gold container. This dataset contains the final inventory status information generated from the Gold Layer business logic and serves as the primary analytical dataset for reporting and dashboard development.

5.3.4 Gold Sales Impact Interfaces

1. D1 Data Extension Implementation

The sample output shows the implementation of the D1 data extension, where the lot_expiry_date attribute was added to the inventory dataset through the gold_inventory_extended view. The generated expiry dates support expiry-based inventory analysis and enable the implementation of Magna Carta Dead Stock (Q3), Magna Carta Expired (Q4), and MC Provision (Q5) business rules.

1SELECT TOP 10 *

2FROM dbo.gold_inventory_extended;

ResultsMessages

View

TableChart

Export results

Search

snapshot_date	material_id	quantity_on_hand	warehouse_loc...	lot_expiry_date
2025-01-31	MAT001	2000	WH-A	2025-06-01
2025-01-31	MAT002	600	WH-B	2026-12-31
2025-01-31	MAT003	1360	WH-C	2026-12-31
2025-01-31	MAT004	360	WH-D	2026-12-31
2025-01-31	MAT005	400	WH-E	2025-06-01
2025-01-31	MAT006	2430	WH-A	2026-12-31
2025-01-31	MAT007	1950	WH-B	2026-12-31
2025-01-31	MAT008	900	WH-C	2026-12-31
2025-01-31	MAT009	1200	WH-D	2026-12-31
2025-01-31	MAT010	800	WH-E	2026-12-31

2. Inventory Dataset Stored in Azure Storage

This figure shows the inventory_snapshot.csv dataset stored in the Bronze Layer inventory container. The dataset serves as the source inventory data used for the D1 data extension process.

bronze > inventory

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

 Search blobs by prefix (case-sensitive)

Showing all 1 items

☐	Name	Last modified
☐	 [..]	
☒	 inventory_snapshot.csv	6/8/2026, 2:26:41 PM

3. Q6 Unfulfilled Production Orders Analysis

The output of the `gold_unfulfilled_production` view with query compares production material requirements against available inventory stock levels to identify production orders that cannot be fulfilled due to insufficient stock. The result returned no records, indicating that all production orders can currently be fulfilled using the available inventory.

1

SELECT * FROM dbo.gold_unfulfilled_production

Results

Messages

View

Table

Chart

Export results

Search

production_order_id

finished_product

material_id

quantity_consumed

current_stock

shortfall

4.Q7 At-Risk Sales Orders and Affected Customers

The output of the `gold_sales_impact` view shows the analysis traces stockout-risk materials through production orders to identify impacted customer sales orders and affected customers.

The result identified two impacted sales orders:

- SO-045 – Maybank Facilities Management
- SO-047 – YTL Hardware

Affected customers include:

- Maybank Facilities Management
- YTL Hardware

```
1 SELECT *
2 FROM dbo.gold_sales_impact;
```

Results Messages

View Table Chart Export results ▼

Search

sales_order_id	customer_name	finished_product	quantity_order...	required_deliv...	material_id	material_name	current_stock	stockout_risk
SO-045	Maybank Facilit...	Marine Blue Co...	275	2026-01-24	MAT008	Epoxy Resin	1420	1
SO-047	YTL Hardware ...	Epoxy Floor Co...	385	2025-12-31	MAT008	Epoxy Resin	1420	1

5. Gold Layer Output Storage

The Gold Layer output folders stored in the Azure Storage gold container. The folders contain the final analytical outputs generated from the Gold Layer business logic.

The Gold Layer outputs include:

- fact_inventory_status
- gold_inventory_extended
- gold_sales_impact
- gold_unfulfilled_production



Authentication method: Access key ([Switch to Microsoft Entra user account](#))

 Search blobs by prefix (case-sensitive)

Showing all 4 items

☐	Name	Last modified
☐	fact_inventory_status	6/8/2026, 4:11:04 PM
☐	gold_inventory_extended	6/8/2026, 4:24:32 PM
☐	gold_sales_impact	6/8/2026, 4:11:11 PM
☐	gold_unfulfilled_production	6/8/2026, 4:11:16 PM

5.4 Testing

5.4.1 Bronze Layer Testing

Test ID	Test Case	Expected Result	Succeed / Failed
A001	materials.csv ingested	20 rows, 0 null material_ids, 4 categories	Pass
A002	inventory_snapshot.csv ingested	240 rows, 12 distinct months, 0 nulls	Pass
A003	purchase_orders.csv ingested	40 rows, 1 open undelivered PO (PO-002)	Pass
A004	production_orders.csv ingested	336 rows, 10 finished products, 0 null qty	Pass
A005	suppliers.csv ingested	10 rows, 0 null supplier_ids, 3 ratings	Pass
A006	sales_orders.csv ingested	48 rows, 10 Open + 38 Delivered	Pass

5.4.2 Silver Layer Testing

Test ID	Test Case	Expected Result	Succeed / Failed
SL001	Materials Transformation	Cleaned Materials dataset generated successfully	Pass
SL002	Inventory Transformation	Cleaned Inventory dataset generated successfully	Pass
SL003	Purchase Orders Transformation	Cleaned Purchase Orders dataset generated successfully	Pass
SL004	Production Orders Transformation	Cleaned Production Orders dataset generated successfully	Pass
SL005	Suppliers Transformation	Cleaned Suppliers dataset generated successfully	Pass

SL006	Sales Orders Transformation	Cleaned Sales Orders dataset generated successfully	Pass
SL007	Parquet File Generation	Output stored in Silver container	Pass
SL008	Data Accessibility	Parquet files can be accessed using OPENROWSET	Pass

5.4.3 Gold RA/MC Logic Testing

Test ID	Test Case	Expected Result	Succeed/ Failed
GL001	Q1 – Below Reorder	System identifies materials with current stock lower than the reorder level	Pass
GL002	Q2 – Recoverable Assets	System identifies materials where current stock exceeds total 12-month consumption	Pass
GL003	Q3 - MC Dead Stock	System identifies materials with no consumption for at least 9 months and no recorded lot expiry date	Pass
GL004	Q4 - MC Expired	System identifies materials with a lot expiry date earlier than the analysis cutoff date (31 December 2025)	Pass
GL005	Q5 - Total MC Provision	System calculates the total provision amount for materials classified as MC Dead Stock or MC Expired	Pass

5.4.4 Gold Sales Impact Testing

Test ID	Test Case	Expected Result	Succeed / Failed
GS001	D1 expiry date extension generated	240 inventory records processed; 180 records assigned valid lot_expiry_date values	Pass
GS002	Expired inventory validation	Expired inventory records successfully identified	Pass
GS003	Q6 unfulfilled production analysis	Production orders successfully validated against inventory stock levels; no production shortfalls identified	Pass
GS004	Q7 at-risk sales order analysis	2 impacted sales orders identified (SO-045, SO-047)	Pass
GS005	Customer impact identification	2 affected customers identified	Pass
GS006	Gold Layer storage validation	Gold Layer output folders successfully created	Pass

5.4.5 Analytics Engine - SQL Execution of the 7 Business Questions

To transition the processed datasets from the Silver layer into highly optimized analytical models, a dimensional Star Schema was engineered in the Gold curated layer. This architecture organizes data into centralized Fact tables capturing transactional metrics surrounded by descriptive Dimension tables to optimize query retrieval speeds.

```
01_Database_Setup x
Run Undo Publish Query plan Connect to Built-in Use database ppg_curated
1 CREATE DATABASE ppg_curated;
2 GO
3 USE ppg_curated;
4 GO
5 -- External Data Source pointing to Gold container
6 CREATE EXTERNAL DATA SOURCE gold_ds WITH (
7     LOCATION = 'https://sappgg4.dfs.core.windows.net/gold'
8 );
9
10 CREATE EXTERNAL FILE FORMAT csv_ff WITH (
11     FORMAT_TYPE = DELIMITEDTEXT,
12     FORMAT_OPTIONS (
13         FIELD_TERMINATOR = ',',
14         STRING_DELIMITER = '"',
15         FIRSTROW = 2 -- untuk skip column header
16     )
17 );
18
19 -- Parquet file format
20 CREATE EXTERNAL FILE FORMAT parquet_ff WITH (FORMAT_TYPE = PARQUET);
```

A dedicated database catalog named ppg_curated is created, and an external data source gold_ds is configured to point directly to the Azure Data Lake Storage container endpoint at https://sappgg4.dfs.core.windows.net/gold. External file formats for both delimited text (csv_ff) and columnar data (parquet_ff) are explicitly defined to enable direct database queries over the underlying storage files.

Q1 - Below Safe Reorder Levels

Running this query returned an empty result set across all columns (material_id, material_name, category, current_stock, reorder_level, shortfall). This confirms that there are currently zero inventory items breaching active safety buffer thresholds.

Search

material_id	material_name	category	current_stock	reorder_level	shortfall
-------------	---------------	----------	---------------	---------------	-----------

00:00:09 Query executed successfully.

Q2 - Recoverable Assets

The query successfully flagged 17 overstocked items across the facility. The total capital locked in excess inventory equals \$1,735,697.50\$, with major concentrations identified in high-value

components such as MAT009 (Polyurethane Resin, \$418,992.00\$ overstock value) and MAT007 (Acrylic Resin, \$215,339.00\$ overstock value).

	A	B	C	D	E	F	G	H	I	J	K
1	material_id,material_name,category,current_stock,consumption_12m,excess_stock,unit_cost,excess_value_usd										
2	MAT009	Polyurethane Resin	Resin	37410	0	37410	11.2	418992			
3	MAT007	Acrylic Resin	Resin	46680	6050	40630	5.3	215339			
4	MAT010	Vinyl Resin	Resin	25200	0	25200	7.6	191520			
5	MAT017	Metal Can 4L	Packaging	127060	5640	121420	1.4	169988			
6	MAT016	Metal Can 1L	Packaging	197810	9840	187970	0.85	159774.5			
7	MAT012	Toluene	Solvent	76960	1080	75880	1.6	121408			
8	MAT014	Acetone	Solvent	48000	0	48000	2.1	100800			
9	MAT011	Xylene	Solvent	58170	7705	50465	1.8	90837			
10	MAT020	Plastic Pail 5L	Packaging	56240	2800	53440	1.1	58784.000000000001			
11	MAT015	Butanol	Solvent	24030	1080	22950	2.4	55080			
12	MAT019	Cardboard Box	Packaging	102000	0	102000	0.45	45900			
13	MAT005	Chrome Yellow	Pigment	5300	0	5300	6.4	33920			
14	MAT018	Plastic Drum 20L	Packaging	15315	5040	10275	3.2	32880			

Q3 - MC Dead Stocks

Execution defines a severe stagnation event for item MAT019 (Cardbox) within the Packaging category. The ledger is recorded as current_stock of 102,000 units with completely unresolved consumption logs (NULL), triggering an automated dead stock risk provision of 45,900.

Search						
material_id	material_name	category	current_stock	last_consumed...	months_since_L...	mc_provision_...
MAT019	Cardboard Box	Packaging	102000	(NULL)	(NULL)	45900

00:00:09 Query executed successfully.

Q4 - MC Expired

The query detects two expired items under the *Pigment* category:

- MAT001 (Titanium Dioxide): 7,970 units in stock at a unit_cost of 8.5, generating an mc_provision_amount liability of 67,745.
- MAT005 (Chrome Yellow): 5,300 units in stock at a unit_cost of 6.4, generating an mc_provision_amount liability of 33,920.

Search					
material_id	material_name	category	current_stock	unit_cost	mc_provision_amount
MAT001	Titanium Dioxide	Pigment	7970	8.5	67745
MAT005	Chrome Yellow	Pigment	5300	6.4	33920

00:00:09 Query executed successfully.

Q5 - Total MC Provision

The aggregate liability matrix compiles a `dead_stock_provision` of 45,900 and an `expired_provision` of 101,665, resulting in a consolidated balance-sheet asset risk write-down requirement of `total_mc_provision` = 147,565.

Search		
dead_stock_provision	expired_provision	total_mc_provision
45900	101665	147565

00:00:09 Query executed successfully.

Q6 - Unfulfilled Production

Tracks manufacturing resource standstills caused by component deficits.

Q7 - At-Risk Orders

The downstream fulfillment analysis isolates a critical stockout risk on MAT008 (Epoxy Resin, current stock holding at 1,420 units) that directly threatens two major open sales orders:

1. SO-047 for YTL Hardware Centre: Ordering 385 units of Epoxy Floor Coating with an explicit delivery deadline of 2025-12-31.
2. SO-045 for Maybank Facilities Mgmt: Ordering 275 units of Marine Blue Coating with a delivery deadline of 2026-01-24.

Search							
sales_order_id	customer_name	finished_product	quantity_ordered	required_delivery_date	material_id	material_name	current_stock
SO-047	YTL Hardware Centre	Epoxy Floor Coating	385	2025-12-31	MAT008	Epoxy Resin	1420
SO-045	Maybank Facilities Mgmt	Marine Blue Coating	275	2026-01-24	MAT008	Epoxy Resin	1420

00:00:09 Query executed successfully.

5.4.6 Enterprise Power BI Visual Analytics Dashboard

The final delivery interface translates the structured relational tables of the Gold Layer into an interactive business intelligence dashboard using Power BI Desktop. To ensure clear data density, the solution uses a scannable multi-page layout arranged by specific business roles.

Table 5.4.6.1 Purpose explanation of every pages in dashboard

Report Page	Analytical Purpose & Operational Purpose
Executive Overview	Designed for senior management to provide a high-level operational pulse. It captures critical core statistics in prominent KPI cards: a Total MC Provision of \$147.565K, an RA Materials Count of 20, a Below Reorder Count of 20, and 2 At-Risk Orders. It includes an interactive donut chart breaking down materials by classification alongside a bar chart mapping financial provision exposure by material category, showing that Pigment represents the largest liability (\$102K).
Recoverable Assets Analysis	Geared toward procurement strategies and warehouse over-purchasing management (Q2). It presents a detailed Recoverable Assets List data grid tracking current stock volumes alongside an interactive Excess Value by Material treemap. This visual instantly isolates heavy capital constraints across material groups, identifying <i>Resin</i> (specifically <i>Polyurethane Resin</i> at \$418.99K and <i>Acrylic Resin</i> at \$215.34K) as the primary drivers of excess holding value.
Magna Carta Risk Analysis	Tailored for accounting and inventory control specialists to manage aging material blocks (Q3 & Q4). The layout splits provisions into two independent tracking tables and high-level KPI cards: Total Dead Stock Provision (\$45.9K) isolating stagnant components like Cardboard Box (102,000 units), and Total Expired Provision (\$101.665K) flagging passed lot lines like Chrome Yellow (\$33,920) and Titanium Dioxide (\$67,745).

Financial Provision Breakdown Summary	Built primarily for corporate finance auditing functions to track aggregate balance sheet liabilities (Q5). It connects individual material provision ledgers directly to a Provision Breakdown by Category waterfall chart. This layout explicitly outlines the step-by-step cost escalation from Pigment (\$102K) and Packaging (\$46K) up to the cumulative asset risk write-down requirement of \$147,565.
Production & Customer Impact Tracing	Engineered for manufacturing schedulers and customer account managers to manage fulfillment blockages (Q6 & Q7). The view cross-references upstream material deficits directly to active downstream customer demand. It provides an operational check that flags impacted clients and their exact order volumes, identifying a total delivery threat of 660 units split between YTL Hardware Centre (385 units via SO-047) and Maybank Facilities Mgmt (275 units via SO-045) due to upstream Epoxy Resin stockouts.

5.5 Dashboard Explanation

The Executive Summary dashboard is shown below in **Figure 5.5.1** for top level operational monitoring. It includes four main KPIs (Total Magna Carta Provision, Recoverable Asset Materials Count, Below-reorder Alerts and At-risk Orders Count) to give a real-time look at inventory status. This consolidated dashboard combines bar and donut charts for breakdowns by categories and provides instant, zero latency interactivity with Import Mode processing for agile decision making.

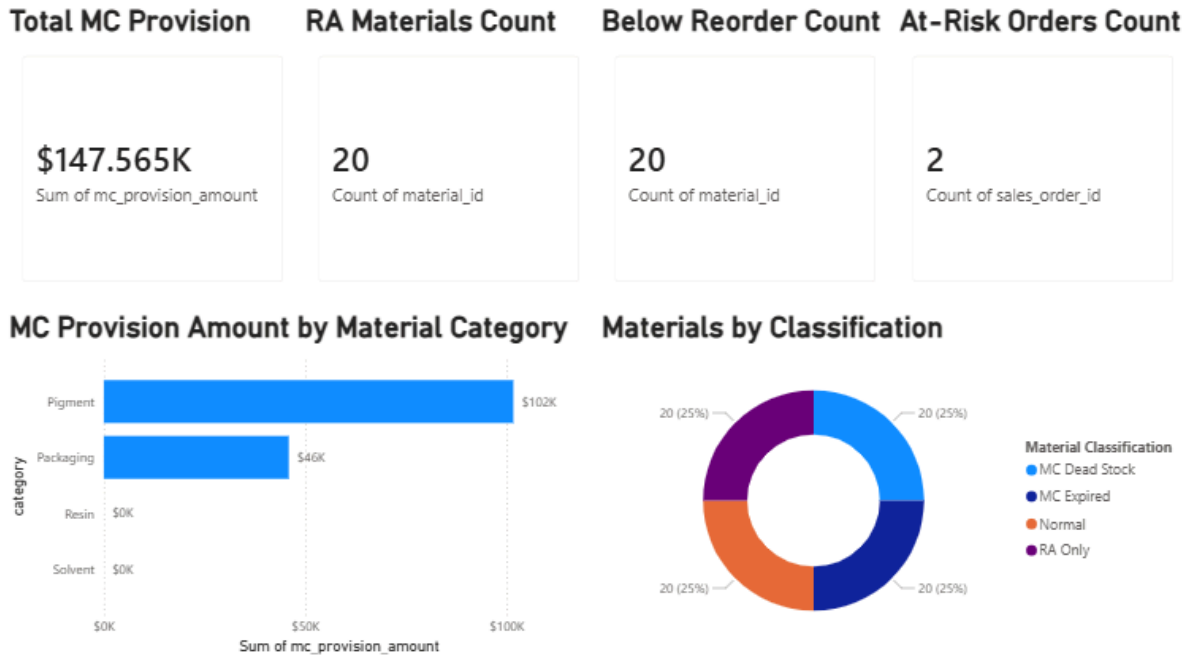


Figure 5.5.1 Real-time Inventory Executive Summary Dashboard

The dashboard in **Figure 5.5.2** below is designed to proactively track inventory shortages, and optimize the replenishment cycle. The detailed matrix populates each material with the precise amount of stock that is out of alignment and the interactive combination chart provides a visual comparison of the stock that is available to the critical reorder levels. The design enables the procurement team to quickly see gaps in stock and make data-informed restocking decisions.

Stock Status List & Shortfall

material_id	category	Sum of current_stock	Sum of reorder_level	Sum of Shortfall
MAT001	Pigment	7970	500	0
MAT002	Pigment	5220	300	0
MAT003	Pigment	5660	200	0
MAT004	Pigment	1920	150	0
MAT005	Pigment	5300	100	0
MAT006	Resin	17610	800	0
MAT007	Resin	46680	600	0
MAT008	Resin	1420	400	0
MAT009	Resin	37410	300	0
MAT010	Resin	25200	200	0
MAT011	Solvent	58170	1000	0
MAT012	Solvent	76960	800	0
MAT013	Solvent	16180	600	0
MAT014	Solvent	48000	400	0
MAT015	Solvent	24030	300	0
MAT016	Packaging	197810	2000	0
MAT017	Packaging	127060	1500	0
MAT018	Packaging	15315	500	0
MAT019	Packaging	102000	1000	0
MAT020	Packaging	56240	800	0
Total		876155	12450	0

Current Stock VS Reorder Level

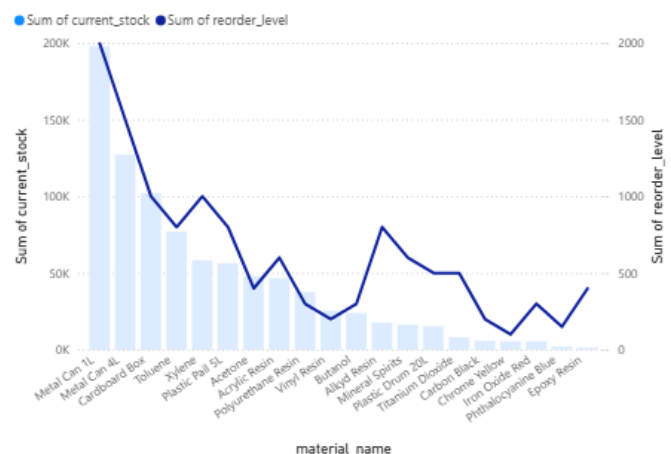


Figure 5.5.2 Real-time Inventory Reorder Alert Dashboard

This dashboard below in **Figure 5.5.3** separates the slow moving inventory that is “recoverable” assets and helps you find “trapped” capital. The detailed matrix then calculates the excesses in stock volume based on consumption volume over a 12-month period, and the accompanying treemap shows the monetary value (USD) of the excesses. The visual hierarchy allows the management team to easily identify high value materials that can be liquidated and/or reallocated to operations.

Recoverable Assets List

Material Name	Category	Sum of current_stock	Sum of consumption_12m	Sum of excess_stock	Sum of excess_v
Oxide Red	Pigment	5220	4860	360	
Black	Pigment	5660	3280	2380	
ral Spirits	Solvent	16180	8200	7980	
l Resin	Resin	17610	11980	5630	
c Drum 20L	Packaging	15315	5040	10275	
me Yellow	Pigment	5300	0	5300	
board Box	Packaging	102000	0	102000	
tol	Solvent	24030	1080	22950	
c Pail 5L	Packaging	56240	2800	53440	\$58,784.000
ie	Solvent	58170	7705	50465	
one	Solvent	48000	0	48000	
me	Solvent	76960	1080	75880	
l Can 1L	Packaging	197810	9840	187970	\$
l Can 4L	Packaging	127060	5640	121420	
Resin	Resin	25200	0	25200	
ic Resin	Resin	46680	6050	40630	
urethane Resin	Resin	37410	0	37410	
		864845	67555	797290	\$1,7

Excess Value by Material

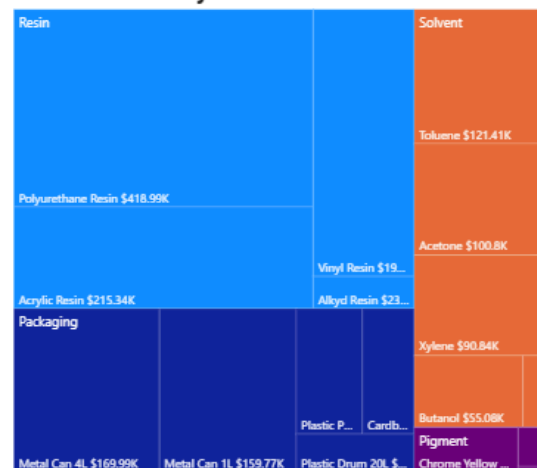


Figure 5.5.3 Recoverable Assets Valuation Dashboard

As mentioned in **Figure 5.5.4**, this dashboard gives a comparative monetary breakdown of written-off inventory, sorted by categories (Dead Stock and Expired materials). The side by side matrix provides a more detailed and granular look at individual items affected, while the stacked bar chart provides a weighted summary of the total amount of provision by provision type. This way, the finance team can immediately identify which areas of inventory are depreciating and make the corresponding adjustments to ensure proper cost allocation.

Total Dead Stock Provision

\$45.9K

Sum of mc_provision_amount

Total Expired Provision

\$101.665K

Sum of mc_provision_amount

Dead Stock Table

material_name	category	Sum of current_stock	Sum of mc_provision_amount
Cardboard Box	Packaging	102000	\$45,900
Total		102000	\$45,900

Expired Table

material_name	category	Sum of current_stock	Sum of mc_provision_amount
Chrome Yellow	Pigment	5300	\$33,920
Titanium Dioxide	Pigment	7970	\$67,745
Total		13270	\$101,665

MC Provision by Category and Type

Material Classification ● MC Dead Stock ● MC Expired

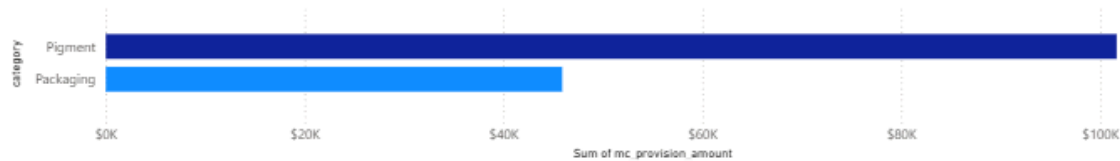


Figure 5.5.4 Magna Carta (Dead Stock & Expired) Dashboard

This dashboard (**Figure 5.5.5**) shows the overall dollars of written off inventory. The total Magna Carta provision is broken down step-by-step in a central waterfall chart, by material category, and the detailed matrix lists the specific dead and expired materials. This transparent financial representation enables stakeholders to track down actual inventory losses item-wise for accurate cost management.

Total MC Provision USD

\$147.565K

Sum of mc_provision_amount

All MC Materials with Individual Provision Amounts

material_name	category	Material Classification	Sum of mc_provision_amount
Titanium Dioxide	Pigment	MC Expired	\$67,745
Cardboard Box	Packaging	MC Dead Stock	\$45,900
Chrome Yellow	Pigment	MC Expired	\$33,920
Total			\$147,565

Provision Breakdown by Category



Figure 5.5.5 Financial Provision Breakdown Dashboard

The final dashboard below, **Figure 5.5.6**, evaluates the impact on customers due to raw materials shortages downstream. The summary matrix consolidates quantities of orders that are at risk, and the detailed table shows the specific orders that are affected. The built-in urgency indicator uses conditional formatting to automatically highlight overdue or soon-to-be-delivered dates in red, allowing logistics and customer service personnel to identify the most important and urgent deliveries and take the initiative in managing client expectations.

At-Risk Customers and Order Quantities

customer_name	Sum of quantity_ordered
Maybank Facilities Mgmt	275
YTL Hardware Centre	385
Total	660

Affected Order List

sales_order_id	customer_name	finished_product	material_name	Sum of quantity_ordered	required_delivery_date
SO-045	Maybank Facilities Mgmt	Marine Blue Coating	Epoxy Resin	275	Saturday, 24 January, 2026
SO-047	YTL Hardware Centre	Epoxy Floor Coating	Epoxy Resin	385	Wednesday, 31 December, 2025
Total				660	

Figure 5.5.6 Stockout Risk and Customer Impact Dashboard

5.6 Chapter Summary

This chapter explains how the inventory risk management system for PPG industries was built using Microsoft Azure. The system uses a three-layer setup to process information. First, raw data from six sources is collected and saved in the Bronze Layer. Next, the Silver Layer cleans and organizes this data so it is easier to use. Finally, the Gold Layer applies specific business rules to the data, such as tracking expiration dates and finding out how long stock might affect production and customer orders.

After building the system, the team tested it thoroughly to make sure everything worked correctly. They checked each step of the process: making sure the raw data was collected right, verifying the data was cleaned properly, and confirming the final business calculations were accurate. All the tests passed successfully without any error. This proved that the automated system is reliable and meets all the project's goals.

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